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(54) **HEAT EXCHANGER FOR STIRLING MACHINE, METHOD FOR MANUFACTURING HEAT EXCHANGER, AND STIRLING MACHINE**

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B33Y 80/00 (2015.01)

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(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,984,428 A * 1/1991 Momose F02G 1/043 60/517
2022/0057147 A1 * 2/2022 Djetel-Gothe B22F 3/1115
2022/0213847 A1 7/2022 Mook et al.

FOREIGN PATENT DOCUMENTS

JP 2022-126185 A 8/2022

OTHER PUBLICATIONS

Jan. 29, 2025, European Search Report issued for related EP Application No. 24198829.4.

* cited by examiner

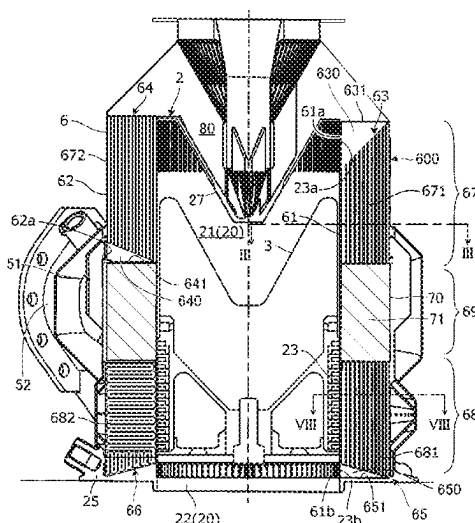
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(57) **ABSTRACT**

A heat exchanger for a Stirling machine that includes an expansion chamber and a compression chamber, the heat exchanger including: a heater; a regenerator; and a cooler, in which the heater includes a first flow channel communicating with the expansion chamber and the regenerator and configured to a working fluid to flow therethrough, and a second flow channel configured to allow a first fluid for exchanging heat with the working fluid to flow therethrough, the cooler includes a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and the heater, the regenerator, and the cooler are continuously connected to each other in series in a flow direction of the working fluid, provided in one piece.

9 Claims, 16 Drawing Sheets



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F28D 19/04; F28D 21/00; F28D 21/0001;
F28D 2021/0019; F25B 9/14; B33Y
80/00
See application file for complete search history.

FIG. 1

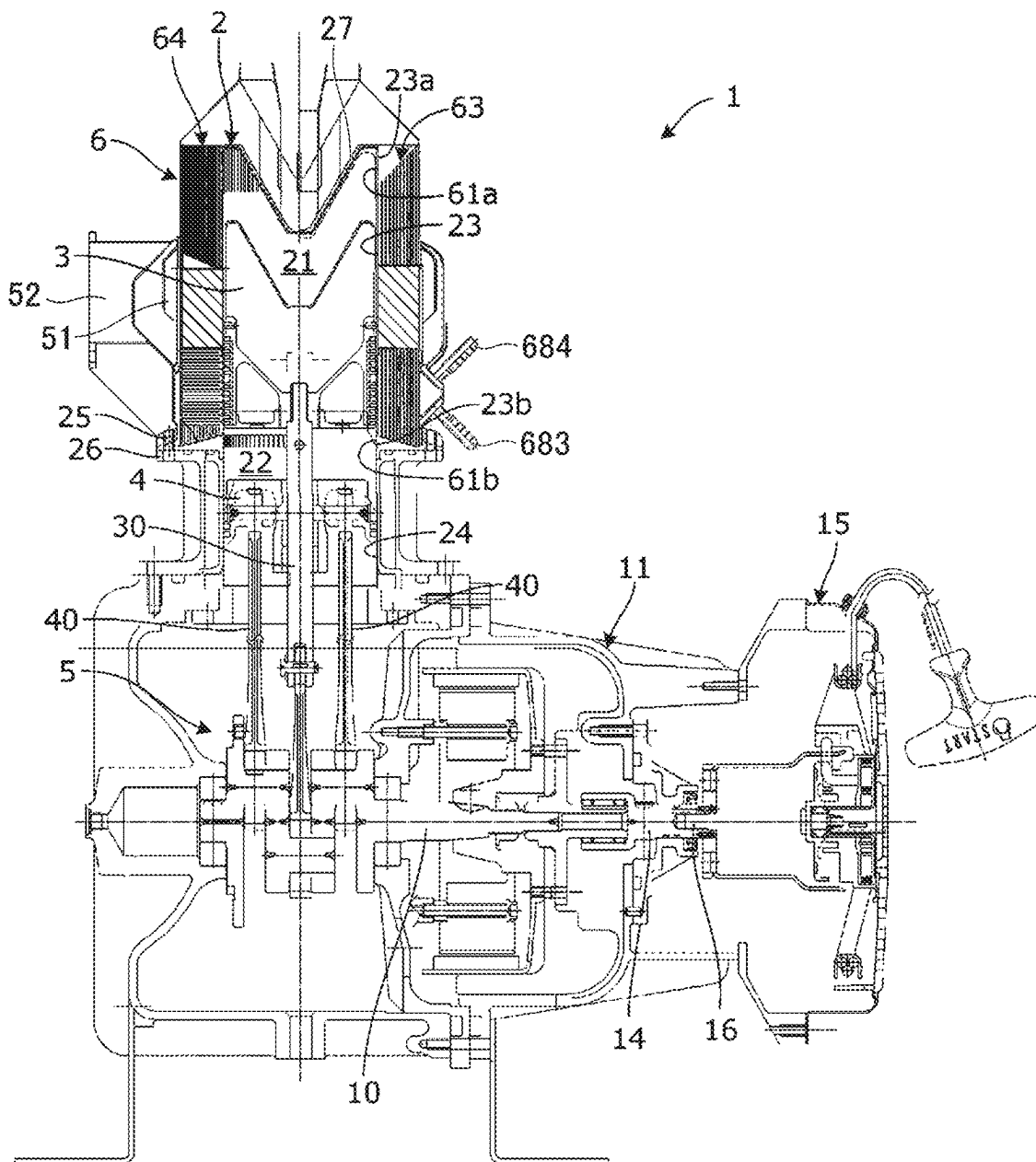


FIG. 2

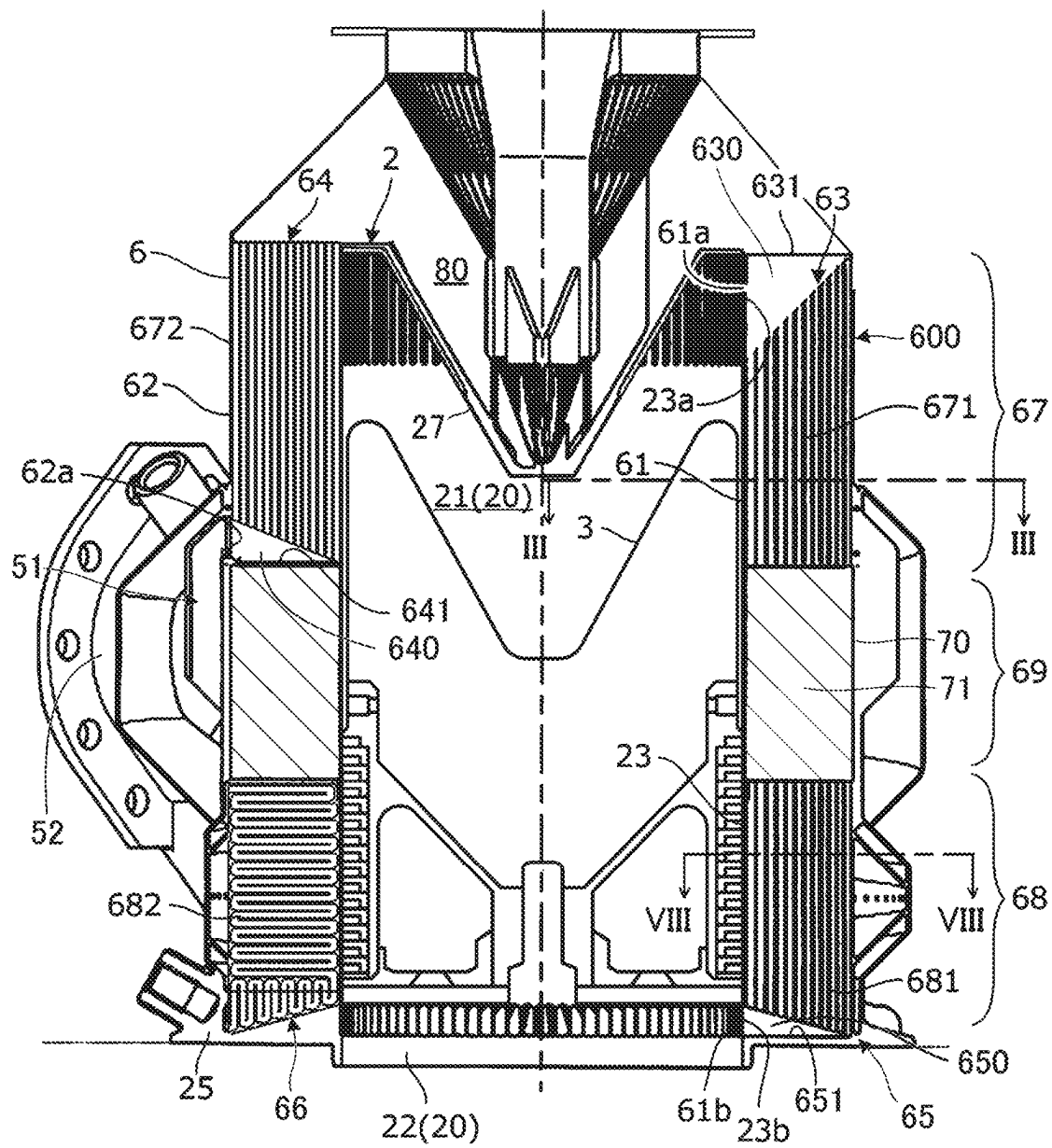


FIG. 3

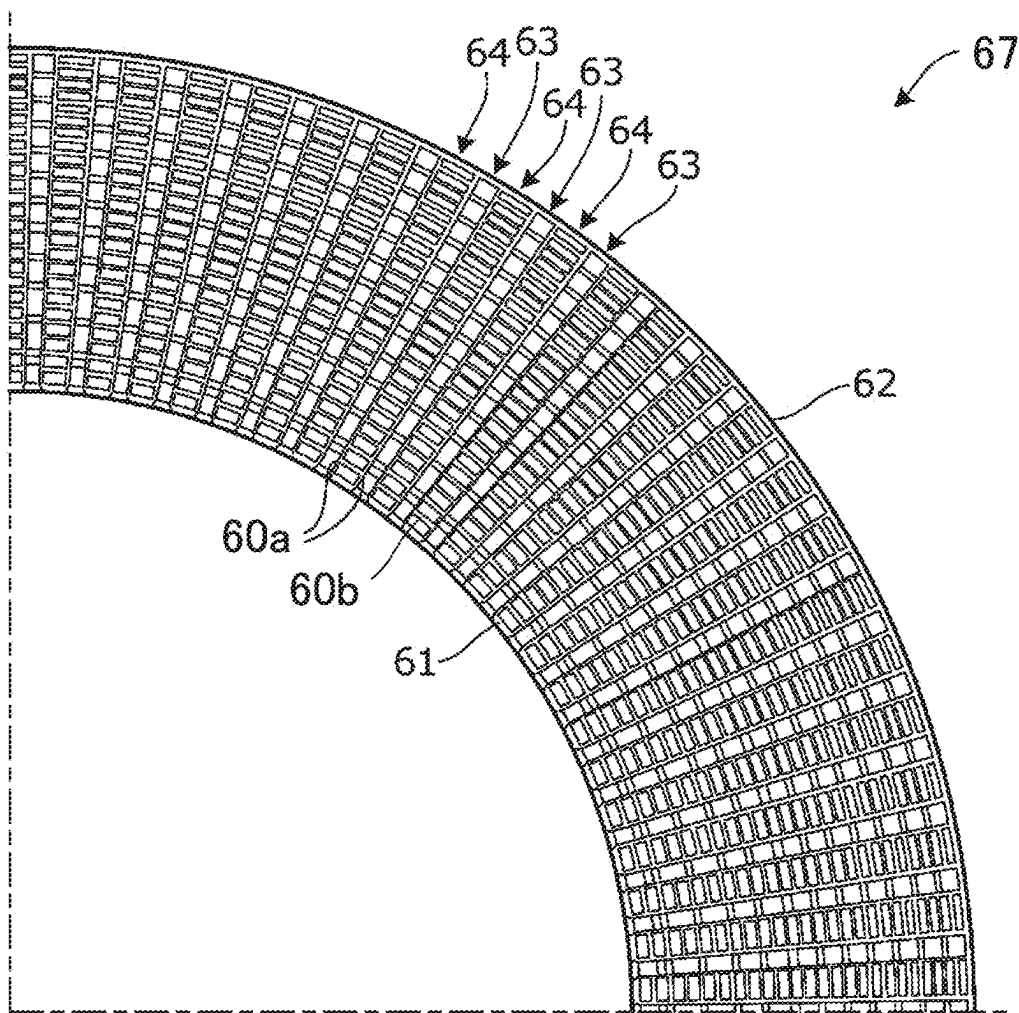


FIG. 4

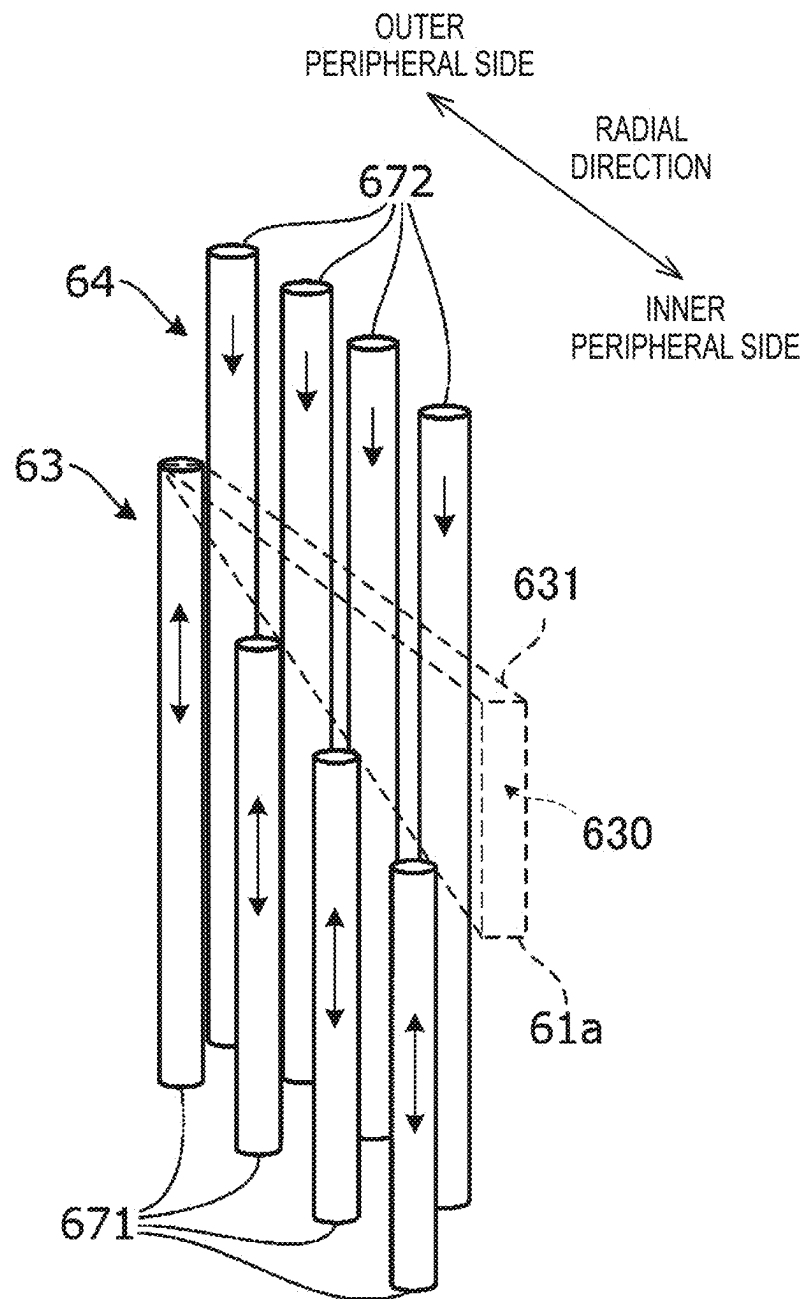


FIG. 5

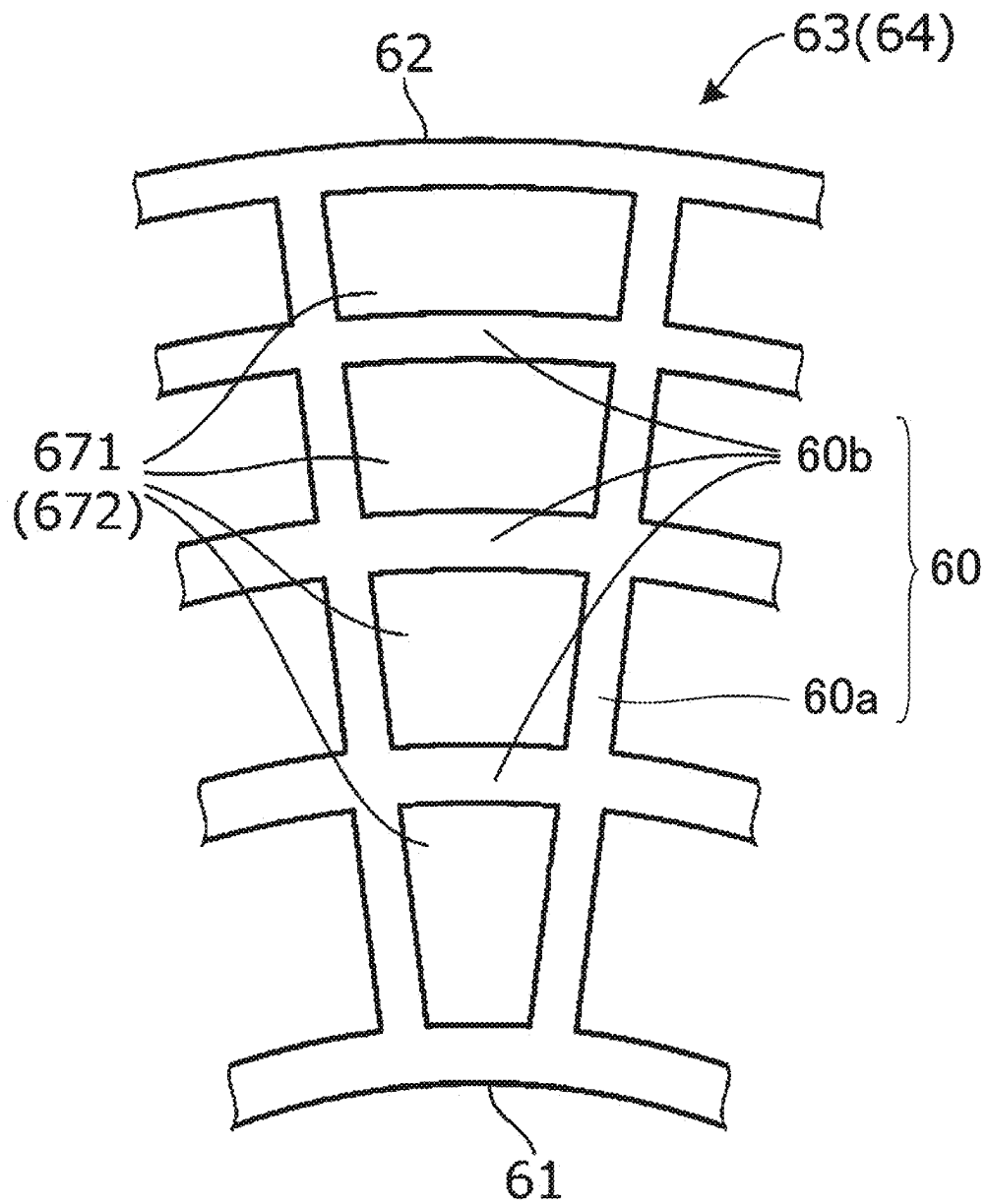


FIG. 6A

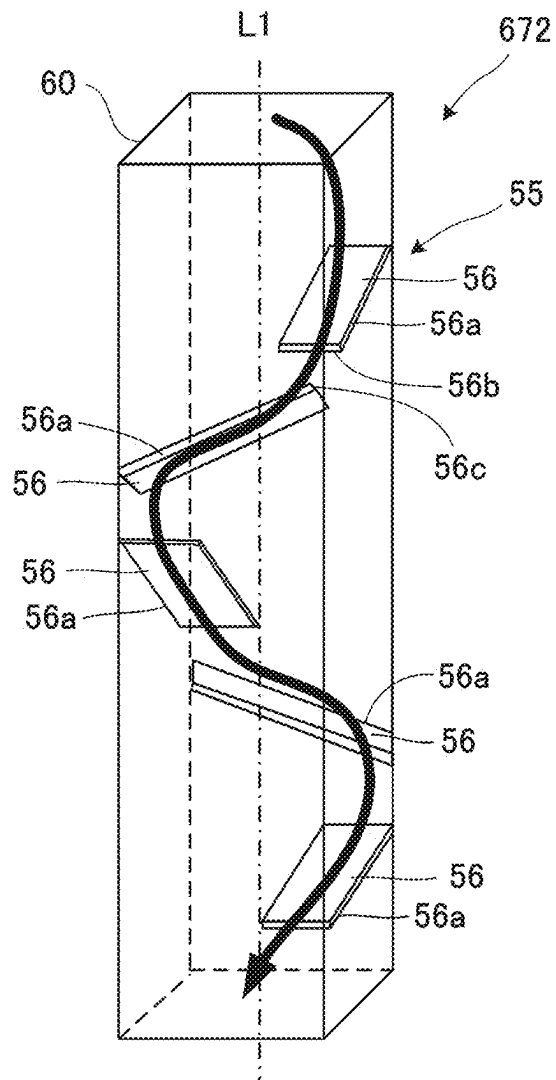


FIG. 6B

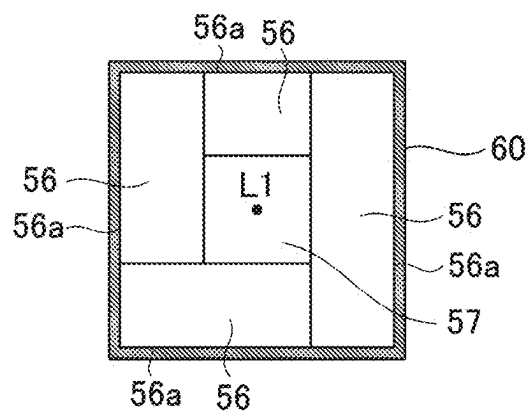


FIG. 7A

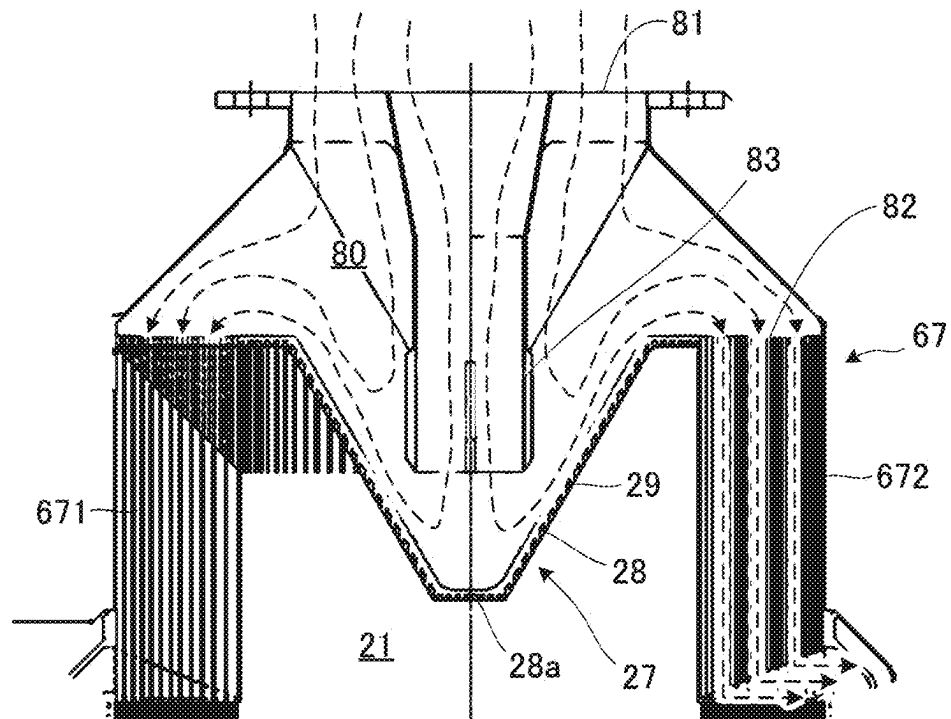


FIG. 7B

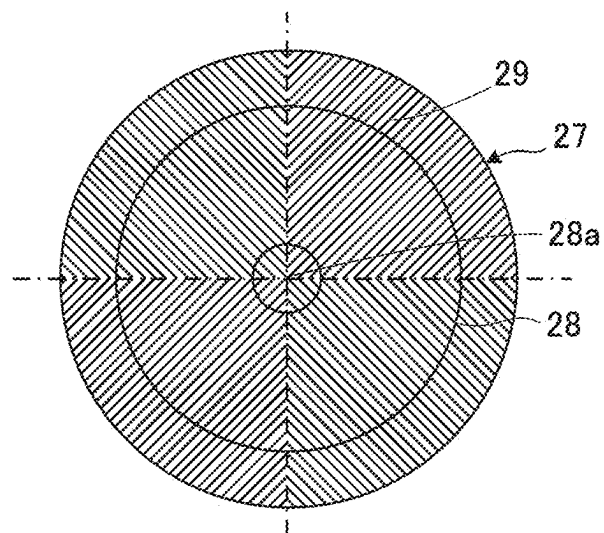


FIG. 8

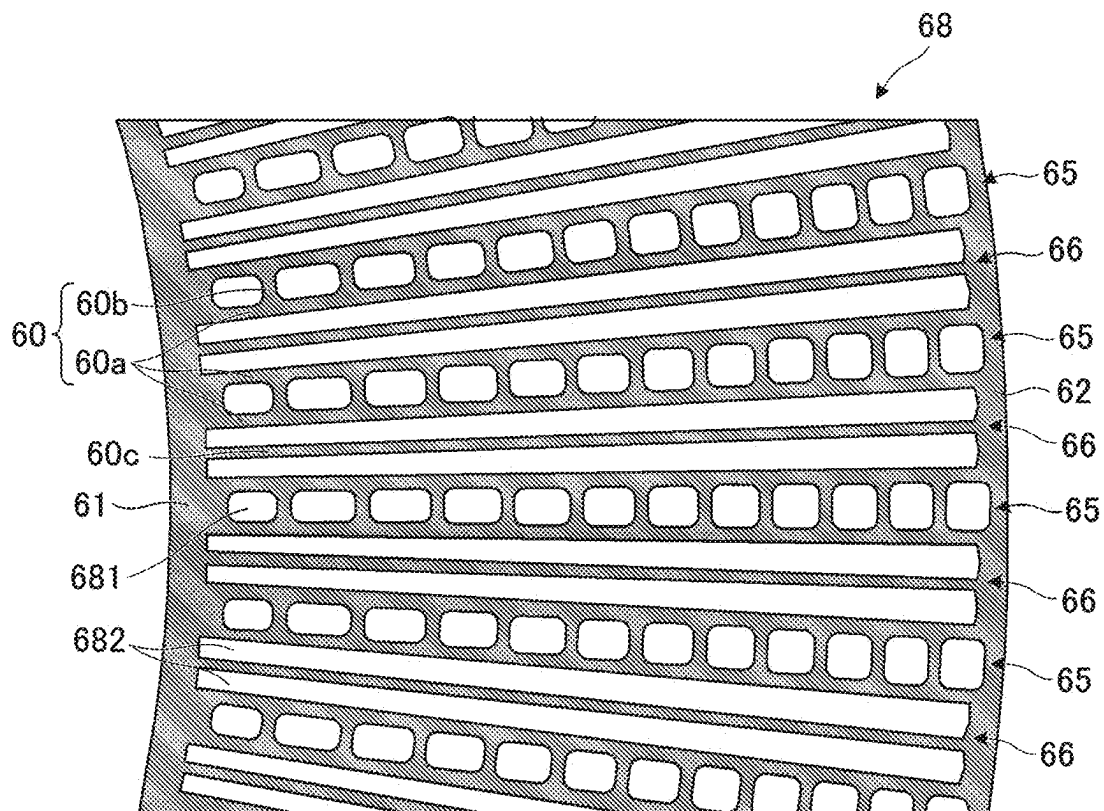
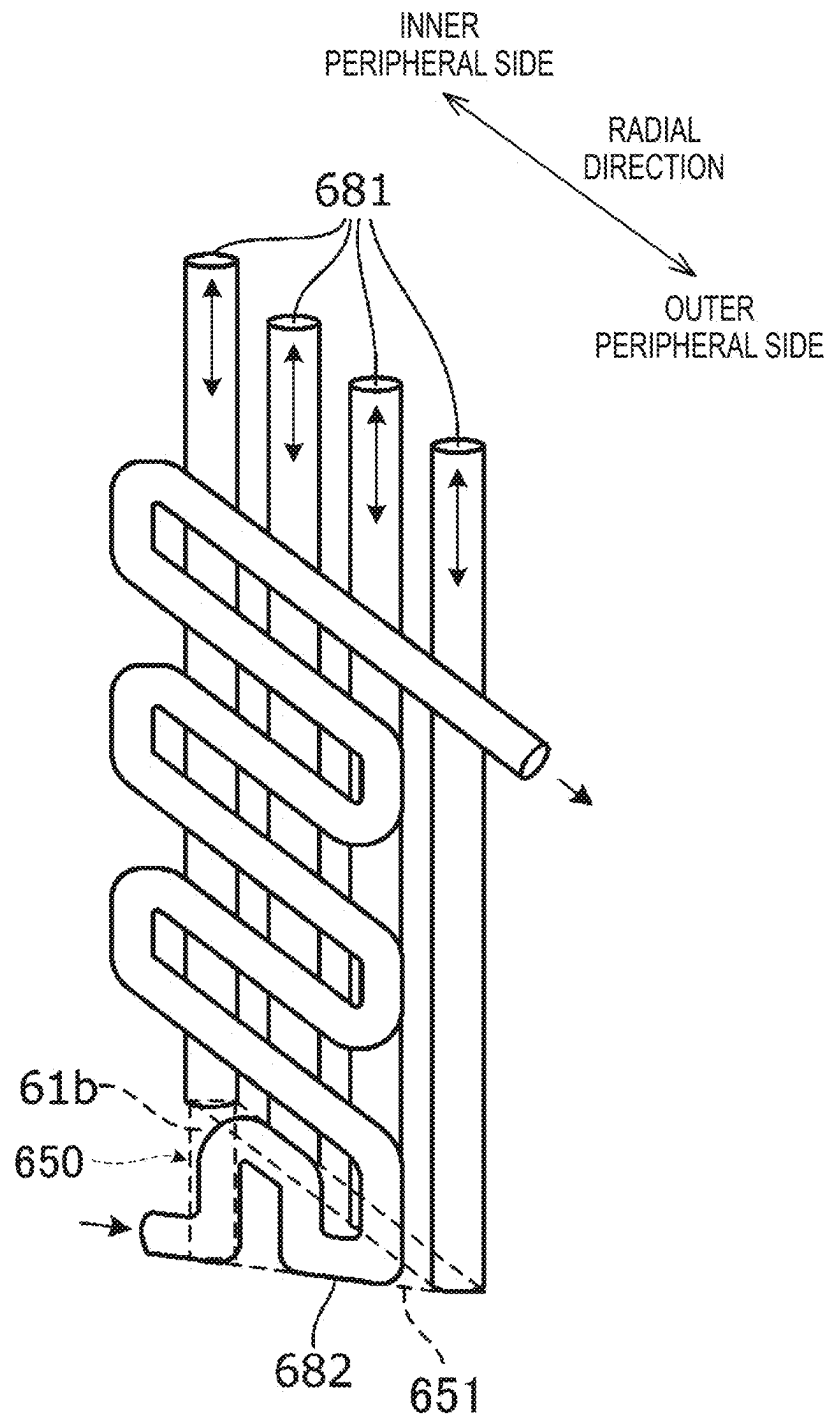
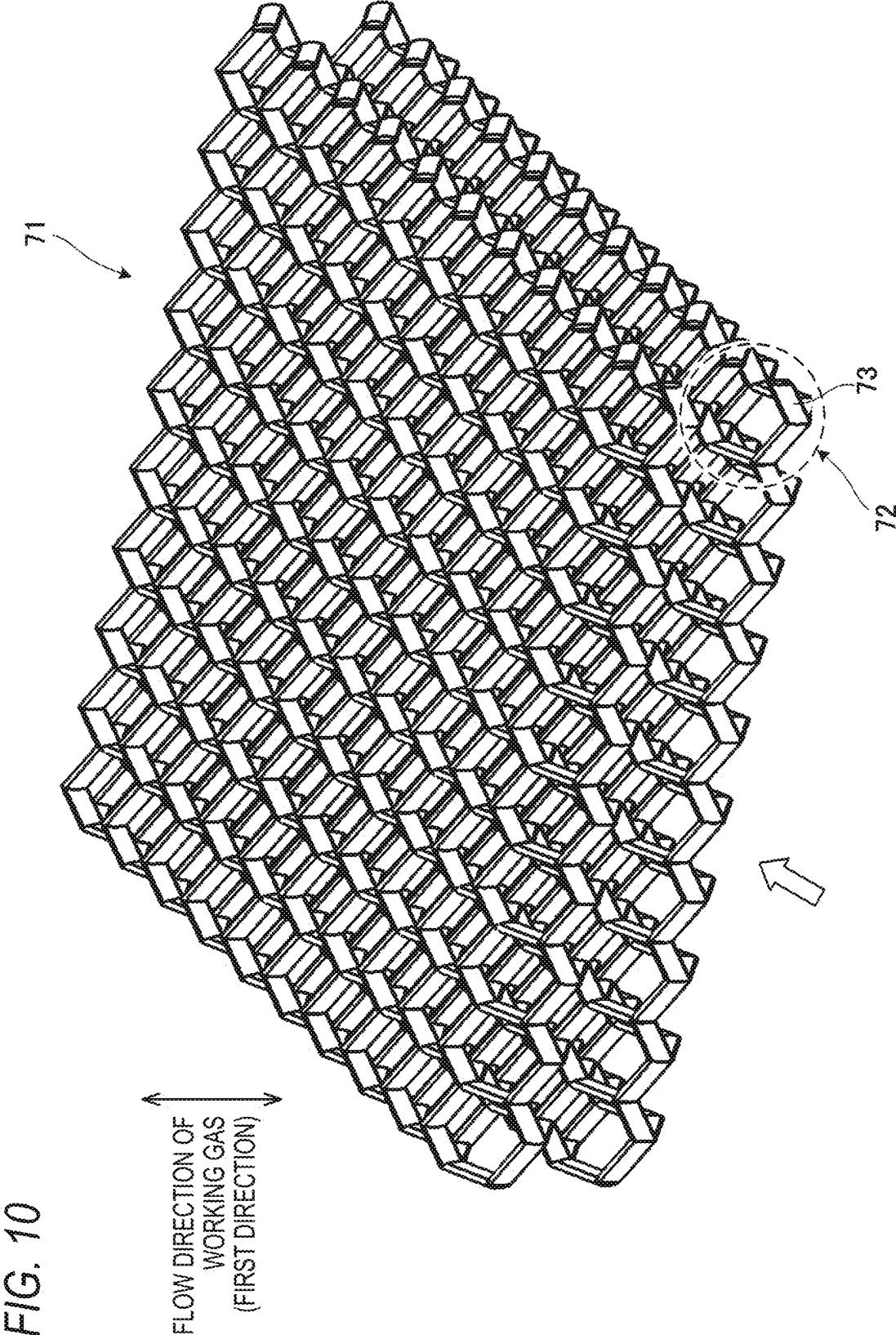


FIG. 9





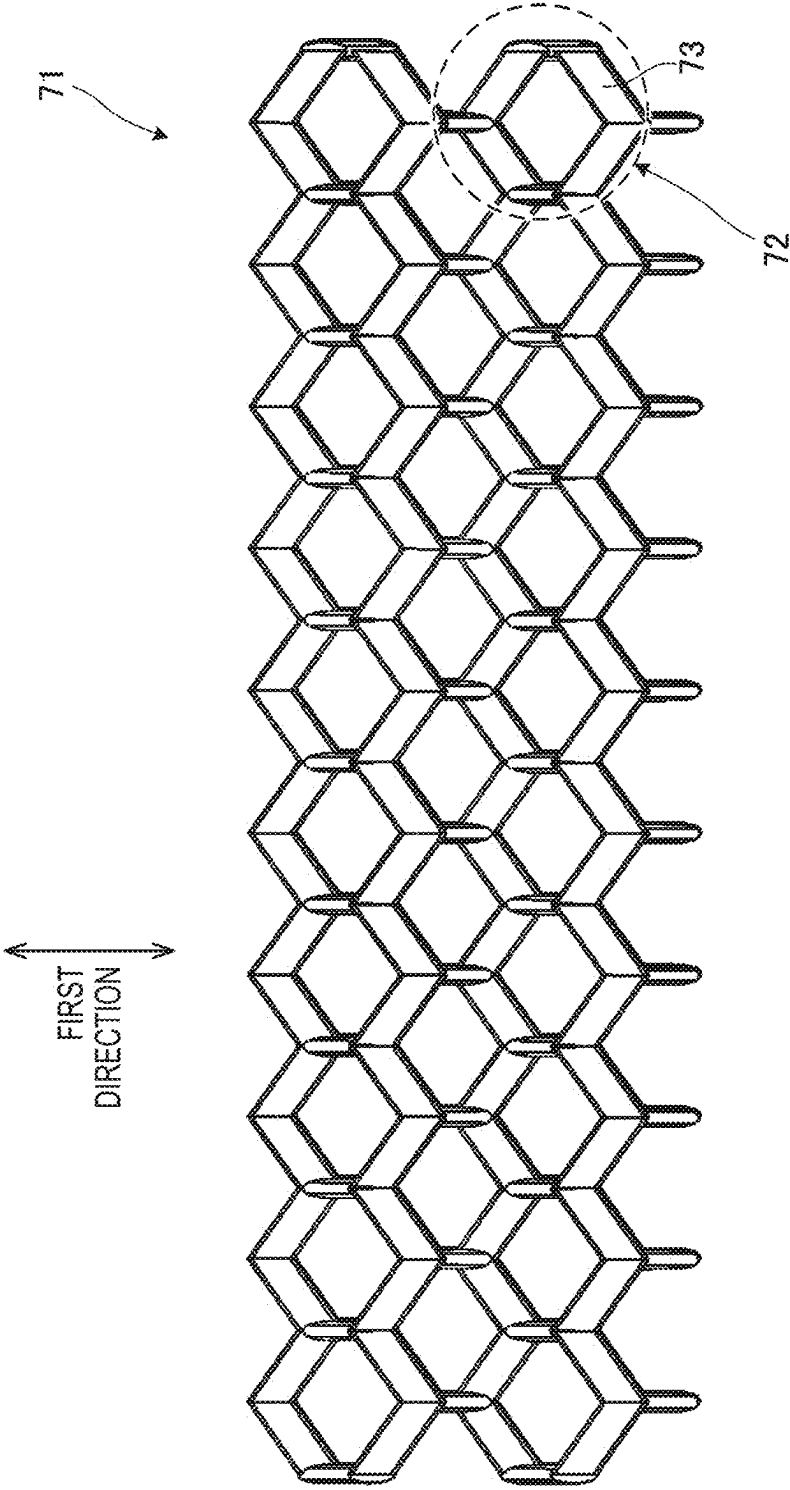


FIG. 11

FIG. 12

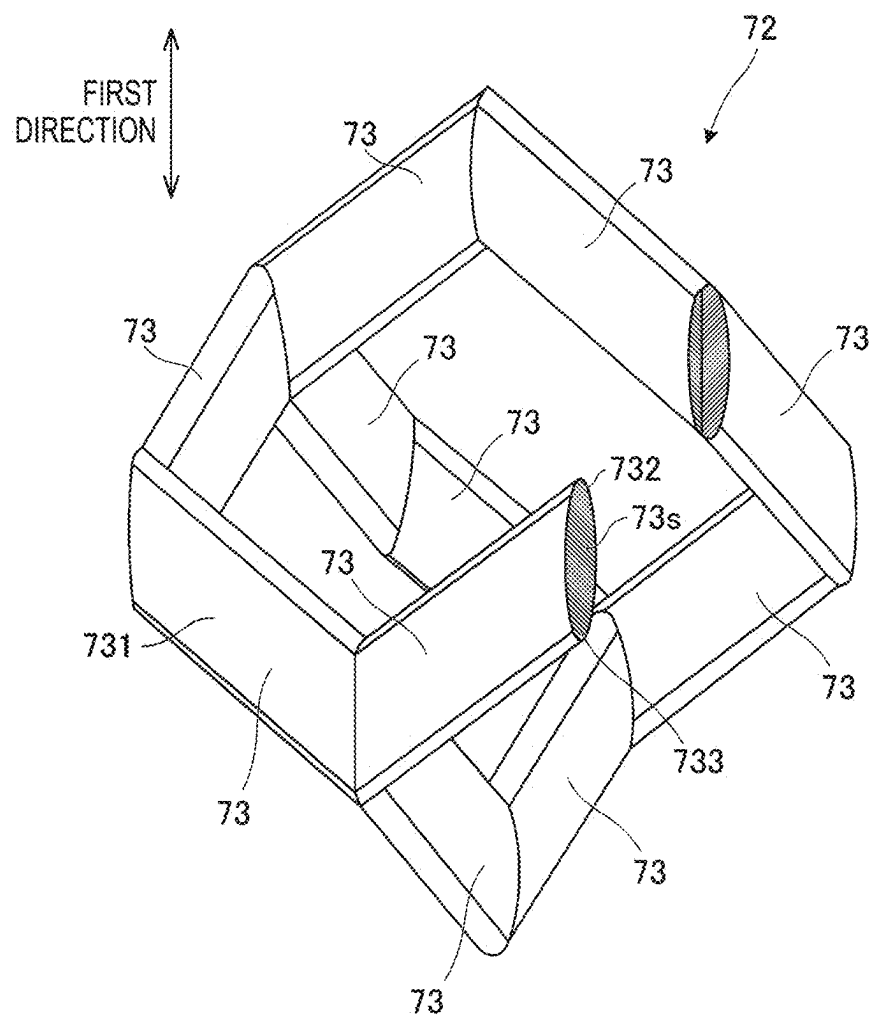


FIG. 13

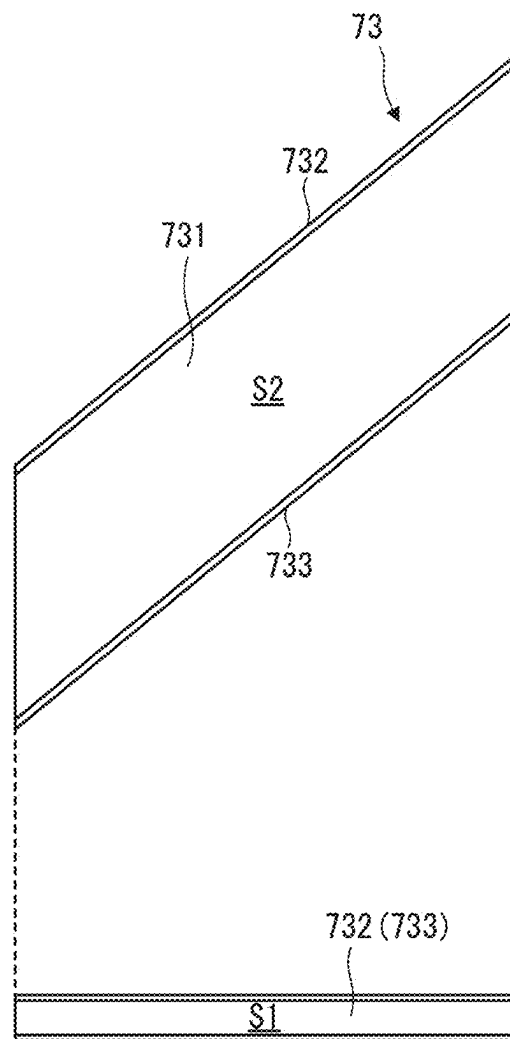


FIG. 14

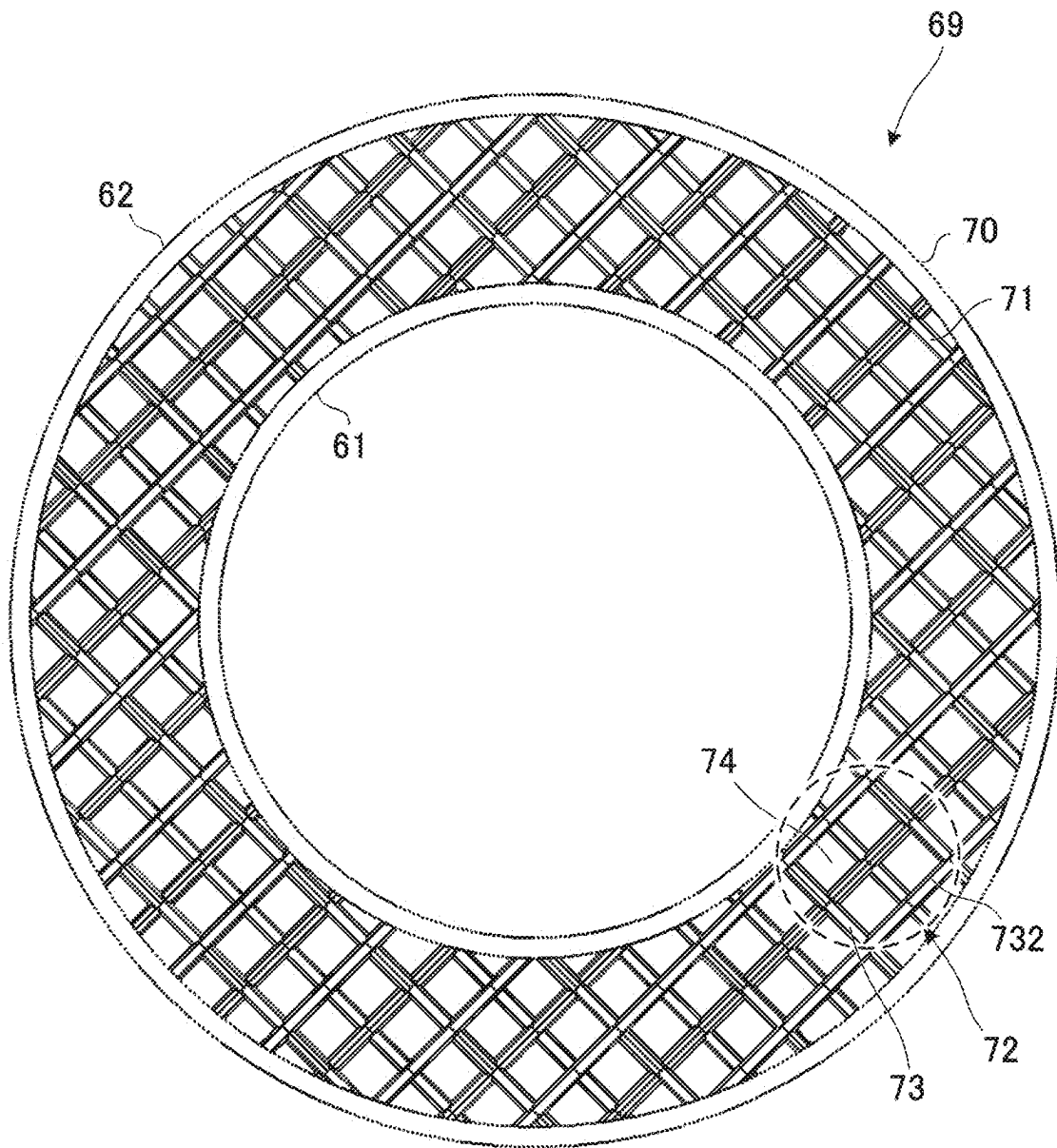


FIG. 15

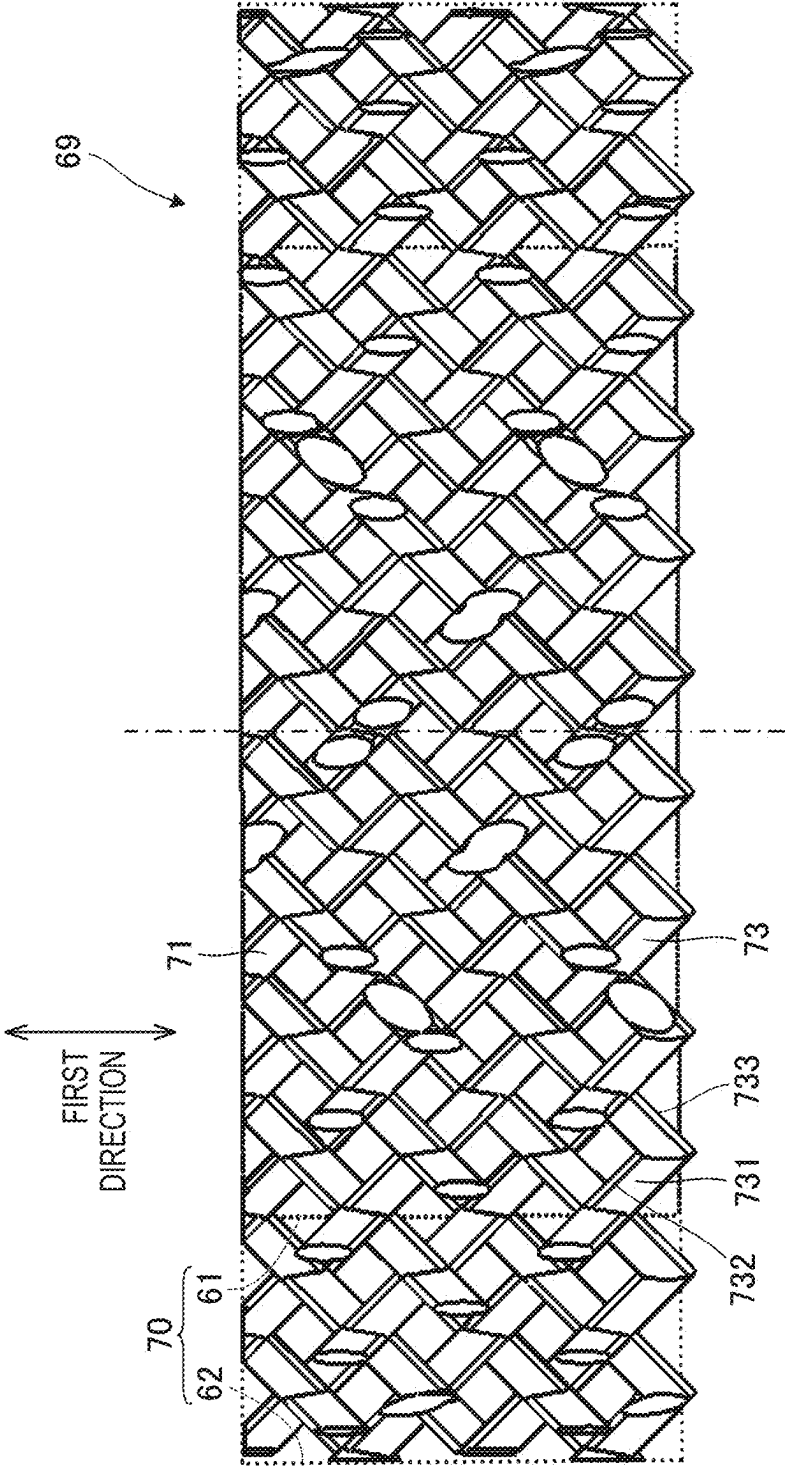
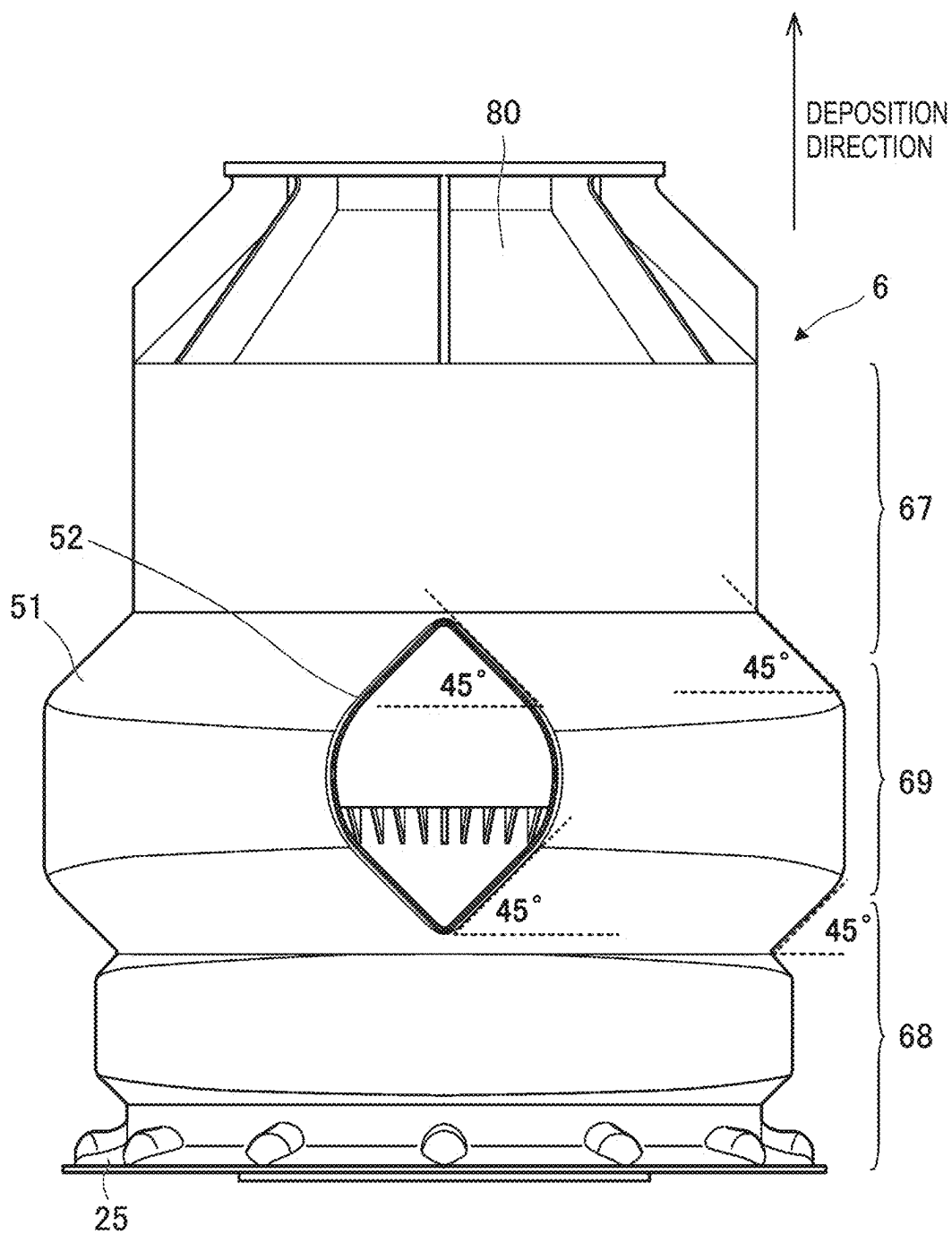


FIG. 16



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HEAT EXCHANGER FOR STIRLING MACHINE, METHOD FOR MANUFACTURING HEAT EXCHANGER, AND STIRLING MACHINE

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to Japanese Patent Application No. 2023-149576 filed on Sep. 14, 2023, the content of which is incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a heat exchanger for a Stirling machine, a method for manufacturing the heat exchanger, and a Stirling machine.

BACKGROUND ART

In recent years, researches and developments have been conducted that contribute to an improvement in energy efficiency in order to allow more people to access affordable, reliable, sustainable, and modern energy.

A Stirling engine is known as a heat engine that performs conversion between heat energy and mechanical power. In the Stirling engine, a working gas is reciprocated between an expansion chamber on a heater side and a compression chamber on a cooler side via a heater, a regenerator, and a cooler to repeat a volume change of the expansion chamber and the compression chamber, thereby performing conversion between heat energy and mechanical power (for example, see JP2022-126185A).

In JP2022-126185A, when manufacturing a heat exchanger including a heater, a regenerator, and a cooler, it is necessary to perform brazing, welding, or the like in order to couple these components, and it takes time and cost in manufacturing the heat exchanger. In addition, when manufacturing the regenerator, a process of stacking metal fibers is necessary, which particularly takes time.

An object of the present invention is to provide a heat exchanger for a Stirling machine, a method for manufacturing the heat exchanger, and a Stirling machine, which can reduce the manufacturing time and the manufacturing cost.

SUMMARY OF INVENTION

According to an aspect of the present invention, there is provided a heat exchanger for a Stirling machine that includes an expansion chamber and a compression chamber, the heat exchanger including:

- a heater;
 - a regenerator; and
 - a cooler, in which
- the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid, and communicate the expansion chamber with the compression chamber,
- the heater includes
- a first flow channel communicating with the expansion chamber and the regenerator and configured to the working fluid to flow therethrough, and
 - a second flow channel configured to allow a first fluid for exchanging heat with the working fluid to flow therethrough,
- the cooler includes

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- a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and
- a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and

the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece.

According to another aspect of the present invention, there is provided a method for manufacturing the heat exchanger, the method including:

- forming the heater, the cooler, and the regenerator in one piece by performing additive manufacturing using a metal powder.

According to still another aspect of the present invention, there is provided a Stirling machine including:

- a heat exchanger including a heater, a regenerator, and a cooler, which are connected in series in a flow direction of a working fluid; and

a cylinder including an expansion chamber connected to the heater and a compression chamber connected to the cooler, in which

the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger,

the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece, and

the heat exchanger and the cylinder are continuously connected to each other, provided in one piece.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view illustrating a Stirling engine 1 as an example of a Stirling machine according to the invention.

FIG. 2 is a cross-sectional view illustrating a configuration of main parts of a heat exchanger 6.

FIG. 3 is a cross-sectional view taken along a line III-III in FIG. 2.

FIG. 4 is a diagram schematically illustrating a working gas flow channel 671 and a heat source fluid flow channel 672 provided in a heater 67.

FIG. 5 is a diagram schematically illustrating a cross section of the working gas flow channel 671 and the heat source fluid flow channel 672 of the heater 67 that are defined and formed by a partition wall 60.

FIG. 6A is a diagram illustrating a spiral flow generation portion 55 provided in the working gas flow channel 671 of the heater 67.

FIG. 6B is a diagram illustrating the working gas flow channel 671 as viewed from a flow direction of a heat source fluid.

FIG. 7A is a diagram illustrating a flow (indicated by broken lines) of a heat source fluid flowing in a heat source fluid introduction chamber 80.

FIG. 7B is a diagram illustrating a ceiling wall 27 as viewed from an expansion chamber 21 side.

FIG. 8 is a cross-sectional view taken along a line VIII-VIII in FIG. 2.

FIG. 9 is a diagram schematically illustrating a working gas flow channel 681 and a cooling fluid flow channel 682 provided in a cooler 68.

FIG. 10 is a perspective view of a heat storage 71.

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FIG. 11 is a side view of the heat storage 71 as viewed in the direction of a white arrow in FIG. 10.

FIG. 12 is a perspective view of a unit structure 72 of the heat storage 71.

FIG. 13 is a diagram illustrating a side element 73 of the heat storage 71 as viewed from a second direction (upper part) and as viewed from a first direction (lower part).

FIG. 14 is a top view of a regenerator 69 in a state where the heat storage 71 is accommodated in a housing 70.

FIG. 15 is a side view of the regenerator 69 in a state where the heat storage 71 is accommodated in the housing 70.

FIG. 16 is a side view of the heat exchanger 6, the heat source fluid introduction chamber 80, and a heat source fluid discharge channel 51 that are formed in one piece.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a Stirling machine of an embodiment of the present invention will be described with reference to the accompanying drawings. Hereinafter, a β -type Stirling engine will be described as an example of the Stirling machine.

[Stirling Machine]

As illustrated in FIG. 1, a Stirling engine 1 includes a cylinder 2 in which a working gas is sealed, a displacer piston 3 and a power piston 4 inserted into the cylinder 2, a piston crank mechanism 5 connected to the displacer piston 3 and the power piston 4, and a heat exchanger 6 that raises and lowers a temperature of the working gas in the cylinder 2. The working gas is, for example, high-pressure air, helium, or hydrogen.

The Stirling engine 1 is, for example, a hermetic engine, and an electric generator 11 that extracts a shaft rotation output of the piston crank mechanism 5 as electric power is disposed inside a crankcase.

The cylinder 2 has a substantially cylindrical shape in which one end thereof is provided with a ceiling wall 27 and the other end thereof is open. In the following description, an axial direction of the cylinder 2 is also referred to as an up-down direction, and the ceiling wall 27 side and the opening side in the axial direction are also referred to as an upper side and a lower side, respectively.

In the cylinder 2, the displacer piston 3 is provided on the upper side, and the power piston 4 is provided on the lower side. An expansion chamber 21 in which the working gas has a high temperature is formed between the displacer piston 3 and the ceiling wall 27, and a compression chamber 22 in which the working gas has a low temperature is formed between the displacer piston 3 and the power piston 4. Hereinafter, the expansion chamber 21 and the compression chamber 22 in the cylinder 2 are also referred to as a working space 20.

The cylinder 2 includes a first cylinder 23 that is positioned on the upper side and a second cylinder 24 that is positioned on the lower side and has the same diameter as the first cylinder 23. The first cylinder 23 and the second cylinder 24 are coaxially connected to each other by flange portions 25 and 26 thereof.

A plurality of first communication portions 23a communicating with a plurality of first inlet-outlets (inlet-outlets) 61a of the heat exchanger 6 respectively, which will be described later, are provided on an outer peripheral surface of the first cylinder 23 on the upper side. The first communication portions 23a are provided in the same number as the first inlet-outlets 61a. Each first communication portion 23a is formed in a slit shape having substantially the same shape

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as the corresponding first inlet-outlet 61a, and is disposed in a manner of facing the corresponding first inlet-outlet 61a. The plurality of first communication portions 23a serve as inlet-outlets to the expansion chamber 21.

A plurality of second communication portions 23b communicating with a plurality of second inlet-outlets (inlet-outlets) 61b of the heat exchanger 6 respectively, which will be described later, are provided on the outer peripheral surface of the first cylinder 23 on the lower side. The second communication portions 23b are provided in the same number as the second inlet-outlets 61b. Each second communication portion 23b is formed in a slit shape having substantially the same shape as the corresponding second inlet-outlet 61b, and is disposed in a manner of facing the corresponding second inlet-outlet 61b. The second communication portions 23b serve as inlet-outlets of the compression chamber 22.

The displacer piston 3 and the power piston 4 are coaxially disposed in the cylinder 2. The displacer piston 3 divides a space defined by the cylinder 2 and the power piston 4 into two in the up-down direction. In other words, the displacer piston 3 constitutes the expansion chamber 21 formed above the displacer piston 3 and the compression chamber 22 formed between the displacer piston 3 and the power piston 4. An internal space of the expansion chamber 21 and the compression chamber 22 (that is, the working space 20) is in communication with an internal space of the heat exchanger 6 via the plurality of first communication portions 23a and the plurality of second communication portions 23b, and has substantially the same pressure as an internal pressure of the heat exchanger 6. Thus, the expansion chamber 21 and the compression chamber 22 communicate with each other via the heat exchanger 6, and the working gas repeatedly reciprocates between the expansion chamber 21 and the compression chamber 22 through the heat exchanger 6 by the movement of the displacer piston 3 described later.

The displacer piston 3 and the power piston 4 are connected to a crankshaft 10 via different connecting rods 30 and 40 respectively such that the power piston 4 has a phase of moving in the lead by 90 degrees (a phase difference of 90 degrees), and the power piston 4 outputs a rotational force to the piston crank mechanism 5 via the connecting rod 40. The crankshaft 10 is coupled to an output shaft 16 of a recoil starter 15 via a coupling 14.

The displacer piston 3 is used to move the working gas in the cylinder 2, and moves the working gas in the cylinder 2 to change a volume ratio between the expansion chamber 21 and the compression chamber 22. For example, when a large amount of high-temperature working gas exists in the expansion chamber 21, the internal pressure of the cylinder 2 is high, and when a large amount of low-temperature working gas exists in the compression chamber 22, the internal pressure of the cylinder 2 is low. In accordance with timings of this pressure fluctuation, when the pressure is high, the power piston 4 is pushed down from near the top dead center to obtain a shaft rotational force to be output to the piston crank mechanism 5, and when the pressure is low, the power piston 4 is pushed up by the piston crank mechanism 5. By repeating such an operation, a continuous rotational force can be obtained.

As described above, the Stirling engine 1 constitutes a Stirling cycle in which the working gas is moved between the expansion chamber 21 and the compression chamber 22 via the heat exchanger 6 to repeat the volume change of the

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expansion chamber 21 and the compression chamber 22, thereby performing conversion between heat and mechanical power.

[Heat Exchanger]

As illustrated in FIG. 2, the heat exchanger 6 is provided in a double cylindrical shape extending in the axial direction (that is, the up-down direction) along the outer peripheral surface of the first cylinder 23, and has an inner peripheral wall 61 provided on an inner peripheral side and an outer peripheral wall 62 provided on an outer peripheral side.

The heat exchanger 6 includes a heater 67 that is provided on the expansion chamber 21 side and performs heat exchange between the working gas and a heat source fluid, a cooler 68 that is provided on the compression chamber 22 side and performs heat exchange between the working gas and a cooling fluid, and a regenerator 69 that is disposed between the heater 67 and the cooler 68 and is capable of storing heat of the working gas. For example, the heat source fluid is a high-temperature gas (for example, high-temperature air), and the cooling fluid is a low-temperature liquid (for example, cooling water).

The heater 67, the regenerator 69, and the cooler 68 are disposed between the inner peripheral wall 61 and the outer peripheral wall 62, and are connected in series in the axial direction of the cylinder 2.

The heater 67, the regenerator 69, and the cooler 68 are provided with a communication path 600 through which the expansion chamber 21 and the compression chamber 22 communicate with each other and through which the working gas is circulated. The communication path 600 includes a plurality of working gas flow channels 671 of the heater 67, an internal space of the regenerator 69, and working gas flow channels 681 of the cooler 68, which will be described later.

[Heater]

As illustrated in FIG. 3, the heater 67 includes a plurality of first partition walls 60a that are provided at predetermined intervals in a circumferential direction and extend in a radial direction, and a plurality of second partition walls 60b that connect facing surfaces of the first partition walls 60a and extend in a flow direction of the working gas and the heat source fluid (that is, the up-down direction).

In the heater 67, working gas circulation layers 63 for circulating the working gas and heat source fluid circulation layers 64 for circulating the heat source fluid are alternately formed in the circumferential direction across the first partition wall 60a. The heater 67 is a plate-type heat exchanger that performs heat exchange between the working gas and the heat source fluid via the first partition wall 60a. In the present embodiment, since the temperature of the heat source fluid is higher than that of the working gas, the heater 67 heats the working gas going into and out of the expansion chamber 21 via the first communication portion 23a of the first cylinder 23.

As illustrated in FIG. 2, the first inlet-outlet 61a is provided at an upper end of the inner peripheral wall 61 where the working gas circulation layer 63 is positioned. The first inlet-outlet 61a is formed in a slit shape having substantially the same shape as the corresponding first communication portion 23a, and is disposed facing the corresponding first communication portion 23a of the first cylinder 23. In the heat exchanger 6, the first inlet-outlet 61a serves as an inlet-outlet to the expansion chamber 21 of the cylinder 2.

An upper end 631 of the working gas circulation layer 63 is closed, and a lower end thereof is open to be in communication with the regenerator 69. The direction in which the

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working gas flowing through the working gas circulation layer 63 flows one time during one rotation of the crankshaft 10 due to the movement of the displacer piston 3 changes to the opposite direction.

An upper end of the heat source fluid circulation layer 64 is open and communicates with a heat source fluid introduction chamber 80. The heat source fluid supplied from the outside is introduced into the heat source fluid circulation layer 64 through the heat source fluid introduction chamber 80. Accordingly, the heat source fluid is introduced into the heat source fluid circulation layer 64 from above. Details of the heat source fluid introduction chamber 80 will be described later.

An outlet 62a is provided in the outer peripheral wall 62 positioned at a lower end of the heat source fluid circulation layer 64. The outlet 62a is formed in a slit shape, is disposed facing the corresponding heat source fluid circulation layer 64, and guides the heat source fluid to a heat source fluid discharge channel 51 provided on the outer side of the heat source fluid circulation layer 64 in the radial direction. An abutting portion 641 is provided at the lower end of the heat source fluid circulation layer 64. The abutting portion 641 is provided along a lower end of the outlet 62a of the outer peripheral wall 62, and guides the heat source fluid to the outlet 62a.

The heat source fluid discharge channel 51 is provided substantially at the center of the heat exchanger 6 in the up-down direction, and extends along the circumferential direction of the heat exchanger 6. The heat source fluid discharge channel 51 is provided with a discharge portion 52, and the heat source fluid guided from each outlet 62a to the heat source fluid discharge channel 51 is discharged from the discharge portion 52 to the outside of the heat exchanger 6.

To describe an inner structure of the working gas circulation layer 63 and the heat source fluid circulation layer 64 in more detail, the heater 67 includes a plurality of working gas flow channels 671 that are provided in the working gas circulation layer 63 and through which the working gas is circulated, and a plurality of heat source fluid flow channels 672 that are provided in the heat source fluid circulation layer 64 and through which the heat source fluid is circulated. Each working gas flow channel 671 and each heat source fluid flow channel 672 are defined and formed by the first partition wall 60a and the second partition wall 60b. The first partition wall 60a and the second partition wall 60b that define and form each flow channel may be collectively referred to as a partition wall 60.

FIG. 4 is a diagram schematically illustrating the working gas flow channel 671 and the heat source fluid flow channel 672, and illustration of the partition wall 60 is omitted in FIG. 4. The working gas flow channel 671 extends in the up-down direction, and an upper end thereof communicates with the expansion chamber 21 via a space 630 and the first inlet-outlet 61a described later, and a lower end thereof communicates with the regenerator 69. In each working gas circulation layer 63, a plurality of working gas flow channels 671 are disposed in parallel along the radial direction. The working gas flows in a counter flow to the heat source fluid when flowing from the lower end to the upper end of the working gas flow channel 671, and the working gas flows in a parallel flow with the heat source fluid when flowing from the upper end to the lower end.

The upper end of the working gas flow channel 671 is positioned farther from the closed upper end 631 of the working gas circulation layer 63 as the working gas flow channel 671 is positioned closer to the first inlet-outlet 61a

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(that is, on the inner peripheral side). That is, the working gas discharged from the working gas flow channel **671** positioned closer to the first inlet-outlet **61a** is discharged at a position farther from the upper end **631** of the working gas circulation layer **63**.

In each working gas circulation layer **63** of the heater **67**, the space **630** defined by the upper end **631** and the upper ends of the plurality of working gas flow channels **671** has a substantially triangular shape in a side view. The space **630** is a portion where the flow of the working gas flowing out of the working gas flow channel **671** is bent toward the first inlet-outlet **61a**. The space **630** has a shape in which a flow channel cross section thereof gradually increases from the outer peripheral side toward the inner peripheral side, and therefore, flow velocities at respective positions are equal and a pressure loss can be minimized. By setting a maximum area of the first inlet-outlet **61a** to be equal to the sum of cross-sectional areas of the plurality of working gas flow channels **671**, the pressure loss of the heat exchanger **6** including the first inlet-outlet **61a** can be minimized.

FIG. **5** is a diagram schematically illustrating a plurality of working gas flow channels **671** provided in each working gas circulation layer **63**. As illustrated in FIG. **5**, the plurality of working gas flow channels **671** are defined and formed by the first partition wall **60a** and the second partition wall **60b**. At this time, the plurality of working gas flow channels **671** are partitioned by the second partition wall **60b** so as to have the same equivalent diameter. By making the equivalent diameters of the respective working gas flow channels **671** equal, a pressure loss coefficient and a heat transfer coefficient become equal, and a drift of the working gas flowing through the respective working gas flow channels **671** is eliminated. By eliminating the drift of the working gas flowing through the working gas flow channel **671**, the overall pressure loss is reduced, and the heat transfer performance is improved.

Returning to FIG. **2**, the heat source fluid flow channel **672** extends in the up-down direction of the heat exchanger **6**, and has an upper end communicating with the heat source fluid introduction chamber **80** and a lower end communicating with the heat source fluid discharge channel **51** via a space **640** and the outlet **62a** described later. In each heat source fluid circulation layer **64**, a plurality of heat source fluid flow channels **672** are disposed in parallel along the radial direction.

The lower end of the heat source fluid flow channel **672** is positioned farther from the abutting portion **641** as the heat source fluid flow channel **672** is positioned closer to the outlet **62a** (that is, on the outer peripheral side). That is, the heat source fluid discharged from the heat source fluid flow channel **672** positioned closer to the outlet **62a** is discharged at a position farther from the abutting portion **641**.

In each heat source fluid circulation layer **64**, the space **640** defined by the abutting portion **641** and the lower ends of the plurality of heat source fluid flow channels **672** has a substantially triangular shape in a side view. The space **640** is a portion where the flow of the heat source fluid flowing out of the heat source fluid flow channel **672** is bent toward the outlet **62a**. The space **640** has a shape in which a flow channel cross section thereof gradually increases from the inner peripheral side toward the outer peripheral side, and therefore, flow velocities at respective positions are equal and a pressure loss can be minimized.

As indicated by the reference signs in parentheses in FIG. **5**, a plurality of heat source fluid flow channels **672** provided in each heat source fluid circulation layer **64** are also defined

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and formed by the first partition wall **60a** and the second partition wall **60b** similarly to the working gas flow channels **671**.

In the present embodiment, the heat conductivity of the heat source fluid is lower than the heat conductivity of the working gas, and the heat transfer coefficient in a flow in which the heat source fluid is circulated through the heat source fluid flow channel **672** is smaller than the heat transfer coefficient in a flow in which the working gas is circulated through the working gas flow channel **671**. In other words, the heat resistance according to the heat transfer of the heat source fluid (hereinafter, also simply referred to as the heat resistance on the heat source fluid side) is larger.

As illustrated in FIGS. **6A** and **6B**, the heat source fluid flow channel **672** is provided with a spiral flow generation portion **55** that generates a spiral flow (indicated by a solid arrow in FIG. **6A**) along the flow direction of the heat source fluid (that is, a downward direction). Accordingly, a tornado-like vortex is generated along the flow direction of the heat source fluid, and the heat source fluid flowing through the heat source fluid flow channel **672** is stirred. Therefore, the heat resistance on the heat source fluid side is reduced, and the heat transfer performance of the working gas and the heat source fluid via the partition wall **60** is improved.

Specifically, the spiral flow generation portion **55** includes a plurality of fins **56** erected from the partition wall **60**. The plurality of fins **56** are disposed at predetermined intervals along the flow direction of the heat source fluid, and are spirally provided around a central axis **L1** of the flow channel cross section of the heat source fluid flow channel **672**. FIGS. **6A** and **6B** illustrate an example in which the flow channel cross section of the heat source fluid flow channel **672** has a rectangular shape, and the plurality of fins **56** are spirally provided at intervals of 90 degrees in the circumferential direction along the central axis **L1**. The flow channel cross section of the heat source fluid flow channel **672** may be any shape. The angular interval in the circumferential direction of the plurality of fins **56** is not limited to 90 degrees. It is sufficient that the fins are provided spirally at angular intervals of 30 degrees or greater and 180 degrees or smaller in the circumferential direction.

Each fin **56** has a base end **56a** connected to the partition wall **60**. The base end **56a** is connected to the partition wall **60** in a manner of inclining at a predetermined angle with respect to a plane perpendicular to the central axis **L1** so as to generate a spiral flow of the heat source fluid. As compared with a case where the fins **56** are not provided, the pressure loss of the heat source fluid flowing through the heat source fluid flow channel **672** is increased due to provision of the fins **56**, but the degree of increase in the pressure loss can be reduced by connecting the fins **56** to the partition wall **60** in a manner of inclining at a predetermined angle. Even when a foreign matter enters the heat source fluid flow channel **672** together with the heat source fluid, the foreign matter is less likely to accumulate on the fins **56**, and thus it is possible to prevent clogging of the heat source fluid flow channel **672**.

The fins **56** adjacent to each other in the flow direction of the heat source fluid are provided at positions not overlapping each other in the flow direction. In other words, a downstream end **56b** of the fin **56** positioned upstream is positioned upstream of an upstream end **56c** of the fin **56** positioned downstream. Accordingly, the degree of increase in the pressure loss when the heat source fluid flows through the heat source fluid flow channel **672** can be further

reduced. Further, it is possible to further prevent clogging of the heat source fluid flow channel 672 involving entry of a foreign matter.

As illustrated in FIG. 6B, the heat source fluid flow channel 672 includes a pass-through region 57, in which the fins 56 are not provided, as viewed from the flow direction of the heat source fluid. In the present embodiment, as viewed from the flow direction of the heat source fluid, a region including the central axis L1 of the heat source fluid flow channel 672 is the pass-through region 57. Accordingly, the degree of increase in the pressure loss when the heat source fluid flows through the heat source fluid flow channel 672 can be further reduced.

As described above, when the spiral flow generation portion 55 is provided in the flow channel, the heat transfer coefficient increases and the heat resistance decreases. On the other hand, as compared with a case where the spiral flow generation portion 55 is not provided, the pressure loss of the fluid increases, and the output performance of the Stirling engine 1 may be affected. Accordingly, it is preferable to determine whether to provide the spiral flow generation portion 55 in the flow channel in consideration of the balance between the effect of the decrease in the heat resistance and the influence of the increase in the pressure loss due to providing the spiral flow generation portion 55 in the flow channel.

Among the working gas, the heat source fluid, and the cooling fluid flowing through the heat exchanger 6, the working gas and the cooling fluid have high heat conductivity. Further, the heat transfer coefficient in a flow in which the working gas is circulated through the working gas flow channel 671 of the heater 67 and the working gas flow channel 681 of the cooler 68 described later, and the heat transfer coefficient in a flow in which the cooling fluid is circulated through the cooling fluid flow channel 682 of the cooler 68 described later are all large. Thus, the heat resistance on the working gas side and the cooling fluid side is small. At this time, the effect of reducing the heat resistance by providing the spiral flow generation portion 55 in the working gas flow channel 671 of the heater 67 and the working gas flow channel 681 and the cooling fluid flow channel 682 of the cooler 68 is relatively insignificant. Accordingly, in consideration of the influence of the increase in the pressure loss, it is preferable that the spiral flow generation portion 55 is not provided in the working gas flow channel 671 of the heater 67 and the working gas flow channel 681 and the cooling fluid flow channel 682 of the cooler 68.

On the other hand, as described above, the heat resistance on the heat source fluid side is large, and the effect of reducing the heat resistance by providing the spiral flow generation portion 55 in the heat source fluid flow channel 672 is significant. Accordingly, even in consideration of the influence of the increase in the pressure loss, it is preferable to improve the heat transfer performance by providing the spiral flow generation portion 55 in the heat source fluid flow channel 672.

Next, the heat source fluid introduction chamber 80 will be described with reference to FIGS. 7A and 7B. The heat source fluid introduction chamber 80 is provided on the outer side (the upper side in the present embodiment) of the expansion chamber 21, and the heat source fluid introduction chamber 80 and the expansion chamber 21 are partitioned by the ceiling wall 27 of the cylinder 2.

The heat source fluid introduction chamber 80 includes an introduction port 81 through which the heat source fluid is introduced from the outside into the heat source fluid

introduction chamber 80, and a guide port 82 serving as an inlet of the heat source fluid from the heat source fluid introduction chamber 80 to the heat source fluid flow channel 672. The introduction port 81 is provided in an upper end of the heat source fluid introduction chamber 80, and the guide port 82 is provided in communication with the upper end of the heat source fluid flow channel 672.

Since the heat source fluid introduction chamber 80 and the expansion chamber 21 are partitioned by the ceiling wall 27, the heat exchange between the heat source fluid flowing through the heat source fluid introduction chamber 80 and the working gas flowing through the expansion chamber 21 is performed via the ceiling wall 27. Accordingly, the working gas on the expansion chamber 21 side is subjected to heat exchange with the heat source fluid not only when the working gas flows through the working gas flow channel 671 but also when the working gas is present in the expansion chamber 21, and therefore, a large amount of heat energy can be taken in from the heat source fluid.

The ceiling wall 27 is provided with a plurality of grooves 29 on the expansion chamber 21 side. The plurality of grooves 29 are not a result of surface roughness formed at the time of manufacturing but formed by design. Although FIG. 7B illustrates an example in which the plurality of grooves 29 are provided on the expansion chamber 21 side, the plurality of grooves 29 may be provided on the outer side of the expansion chamber 21, that is, on the heat source fluid introduction chamber 80 side, or may be provided on both the expansion chamber 21 side and the heat source fluid introduction chamber 80 side.

The plurality of grooves 29 are provided in the entire region of the ceiling wall 27 and extend from a center side of the ceiling wall 27 toward an outer peripheral side (the heater 67 side).

Since a surface area of the ceiling wall 27 is increased by providing the plurality of grooves 29 in the ceiling wall 27, the working gas can take more heat energy from the heat source fluid, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall 27 is improved. As a result, the heat efficiency and output performance of the Stirling engine 1 can be improved. The plurality of grooves 29 are not necessarily provided in the entire region of the ceiling wall 27, and may be provided in parts of the ceiling wall 27.

FIG. 7B illustrates an example in which the plurality of grooves 29 linearly extend from the center side to the outer peripheral side, and the present invention is not limited thereto. The plurality of grooves 29 may extend from the center side to the outer peripheral side in, for example, a zigzag shape (a shape in which the extending direction thereof changes in the middle), or may extend from the center side to the outer peripheral side in a curved shape. Further, depending on the position of the ceiling wall 27, the plurality of grooves 29 may be a combination of linear, zigzag, and curved grooves. Thus, more heat energy can be taken in by appropriately changing the shape of the plurality of grooves 29.

The ceiling wall 27 is provided with a substantially conical protruding portion 28 protruding toward the expansion chamber 21. The protruding portion 28 is formed by recessing a portion of the ceiling wall 27 toward the expansion chamber 21, and is provided such that a central axis thereof coincides with the central axis of the cylinder 2. Since the protruding portion 28 is provided on the ceiling wall 27, the surface area of the ceiling wall 27 can be increased as compared with a case where the ceiling wall 27 is formed in a flat plate shape. Accordingly, the heat energy

taken in by the working gas can be further increased, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall 27 is further improved. Further, by providing the protruding portion 28, the pressure resistance of the ceiling wall 27 is also increased.

The heat source fluid introduction chamber 80 is provided with a guide portion 83 that guides the heat source fluid toward an apex part 28a of the protruding portion 28 of the ceiling wall 27. The guide portion 83 extends along the central axis of the protruding portion 28 to a position overlapping the protruding portion 28 in the axial direction of the cylinder 2. As indicated by broken lines in FIG. 7A, the guide portion 83 guides the heat source fluid flowing in from the introduction port 81 to the apex part 28a of the ceiling wall 27. The heat source fluid guided to the apex part 28a by the guide portion 83 flows along the substantially conical protruding portion 28 to the outer side in the radial direction, and is introduced into the heat source fluid flow channel 672 from the guide port 82. Accordingly, the heat source fluid easily flows along the ceiling wall 27, and the heat exchange efficiency between the working gas and the heat source fluid via the ceiling wall 27 can be further improved.

Since the plurality of grooves 29 extend from the apex part 28a of the protruding portion 28 toward the heater 67 side, the heat source fluid can smoothly flow from the apex part 28a toward the heater 67 side.

[Cooler]

As illustrated in FIG. 8, similarly to the heater 67, the cooler 68 includes a plurality of first partition walls 60a that are provided at predetermined intervals in the circumferential direction and extend in the radial direction.

As illustrated in FIGS. 2 and 8, in the cooler 68, working gas circulation layers 65 for circulating the working gas and cooling fluid circulation layers 66 for circulating the cooling fluid are alternately formed in the circumferential direction across the first partition wall 60a. The cooler 68 is a plate-type heat exchanger that performs heat exchange between the working gas and the cooling fluid via the first partition wall 60a. In the present embodiment, since the temperature of the cooling fluid is lower than that of the working gas, the cooler 68 cools the working gas flowing into and out of the compression chamber 22 via the second communication portion 23b of the first cylinder 23.

The second inlet-outlet 61b described above is provided at the lower end of the inner peripheral wall 61 where the working gas circulation layer 65 is positioned. The second inlet-outlet 61b is formed in a slit shape having substantially the same shape as the corresponding second communication portion 23b, and is disposed facing the corresponding second communication portion 23b of the first cylinder 23. In the heat exchanger 6, the second inlet-outlet 61b serves as an inlet-outlet to the compression chamber 22 of the cylinder 2.

A lower end 651 of the working gas circulation layer 65 is closed, and an upper end thereof is open to be in communication with the regenerator 69. The direction in which the working gas flowing through the working gas circulation layer 65 flows one time during one rotation of the crankshaft 10 due to the movement of the displacer piston 3 changes to the opposite direction.

To describe an inner structure of the working gas circulation layer 65 and the cooling fluid circulation layer 66 in more detail, the cooler 68 includes a plurality of working gas flow channels 681 that are provided in the working gas circulation layer 65 and through which the working gas is circulated, and a plurality of cooling fluid flow channels 682 that are provided in the cooling fluid circulation layer 66 and

through which the cooling fluid is circulated. In the present embodiment, two cooling fluid flow channels 682 are provided side by side in the circumferential direction in each cooling fluid circulation layer 66, and are partitioned by a third partition wall 60c extending in the radial direction. Similarly to the working gas flow channels 671 and the heat source fluid flow channels 672, the working gas flow channels 681 are also defined and formed by the first partition wall 60a and the second partition wall 60b. Since the configuration of the working gas flow channel 681 is similar to the configuration of the working gas flow channel 671 provided in the heater 67, a detailed description thereof will be omitted.

FIG. 9 is a diagram schematically illustrating the working gas flow channel 681 and the cooling fluid flow channel 682 provided in the cooler 68. In FIG. 9, the partition wall 60 is not shown.

The working gas flow channel 681 extends in the up-down direction, and a lower end thereof communicates with the compression chamber 22 via a space 650 and the second inlet-outlet 61b described later, and an upper end thereof communicates with the regenerator 69. In each working gas circulation layer 65, a plurality of working gas flow channels 681 are disposed in parallel along the radial direction.

The lower end of the working gas flow channel 681 is positioned farther from the closed lower end 651 of the working gas circulation layer 65 as the working gas flow channel 681 is positioned closer to the second inlet-outlet 61b (that is, on the inner peripheral side). That is, the working gas discharged from the working gas flow channel 681 positioned closer to the second inlet-outlet 61b is discharged at a position farther from the lower end 651 of the working gas circulation layer 65.

Also in the working gas circulation layer 65 of the cooler 68, the space 650 defined by the lower end 651 and the lower ends of the plurality of working gas flow channels 681 has a substantially triangular shape in a side view. Similarly to the plurality of working gas flow channels 671 in each working gas circulation layer 63, the plurality of working gas flow channels 681 in each working gas circulation layer 65 are formed by partitioning a space between a pair of first partition walls 60a by the second partition walls 60b. Therefore, the description of the working gas flow channel 671 is referenced to, and is omitted here.

Each cooling fluid flow channel 682 is formed by folding back a tubular member extending in the radial direction at one end (for example, an end on the outer peripheral side) in the radial direction and then folding back the tubular member at the other end (for example, an end on the inner peripheral side), which is repeated. Each cooling fluid flow channel 682 is formed in a so-called zigzag shape. Each cooling fluid flow channel 682 is formed by bending one flow channel into a folded state instead of being formed with a plurality of linear flow channels, and thus the contact surface of the cooling fluid with the cooling fluid flow channel 682 can be increased.

As illustrated in FIG. 1, the cooling fluid flow channel 682 includes an inlet 683 provided on the lower end side in the up-down direction and an outlet 684 provided on the upper end side, and is configured such that the cooling fluid flows upward as a whole.

When the working gas heated by the heater 67 flows from the upper end side to the lower end side, that is, into the cooler 68, the working gas flowing through the working gas flow channel 681 has a temperature distribution in which the temperature decreases from the upper end side to the lower end side, that is, from the expansion chamber 21 side to the

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compression chamber 22 side, and the cooling fluid flowing through the cooling fluid flow channel 682 has a temperature distribution in which the temperature increases from the lower end side to the upper end side. In other words, when the working gas flows from the upper end side to the lower end side, the working gas and the cooling fluid flow in the working gas flow channel 681 and the cooling fluid flow channel 682 so as to have a counter flow temperature distribution as a whole, and thus high heat exchange efficiency can be achieved in the cooler 68. Further, since the contact area of the working gas and the cooling fluid is increased by the cooling fluid flow channel 682 formed in the zigzag shape, the heat exchange efficiency is further improved in the cooler 68. When the working gas flows from the lower end side to the upper end side, the working gas and the cooling fluid flow in the working gas flow channel 681 and the cooling fluid flow channel 682 so as to have a parallel flow temperature distribution as a whole.

[Regenerator]

When the working gas moves from the high-temperature expansion chamber 21 to the low-temperature compression chamber 22, the regenerator 69 cools the working gas by taking in the heat energy of the working gas (that is, storing heat). Conversely, when the working gas moves from the low-temperature compression chamber 22 to the high-temperature expansion chamber 21, the regenerator 69 heats the working gas by imparting the stored heat energy to the working gas (that is, releasing heat).

As illustrated in FIG. 2, the regenerator 69 includes a heat storage 71 capable of storing heat of the working gas and a housing 70 surrounding the heat storage 71.

The housing 70 includes the inner peripheral wall 61 and the outer peripheral wall 62 described above, and is provided in a double cylindrical shape. The heat storage 71 is accommodated in an internal space of the housing 70.

In the present embodiment, the heat storage 71 and the housing 70 are continuously connected to each other and provided in one piece. Specifically, the regenerator 69 is an additive manufactured body in which metal powders are accumulated, and the heat storage 71 and the housing 70 are provided in one piece by metal additive manufacturing. According to such a configuration, it is possible to adopt a complicated structure having high heat exchange performance as compared with a regenerator in the related art that is formed by integrating a metal mesh, metal foam, or the like as a heat storage into a housing. Further, the number of assembly steps can be reduced.

Hereinafter, a structure of the heat storage 71 will be described with reference to FIGS. 10 to 15. The structure of the heat storage 71 illustrated in FIGS. 10 to 15 is an example, and the structure of the heat storage 71 can take various shapes. In the following description, an up-down direction in the drawings, that is, a flow direction of the working gas is also referred to as a first direction, and a horizontal direction in the drawings, that is, a direction intersecting the first direction is also referred to as a second direction.

The heat storage 71 includes a plurality of side elements 73, and the plurality of side elements 73 are continuously connected to each other in a three-dimensional manner and provided in one piece.

More specifically, as illustrated in FIGS. 10 and 11, the heat storage 71 includes a plurality of unit structures 72 each formed by connecting a plurality of side elements 73 to each other, and the unit structures 72 are continuously provided in the flow direction of the working gas and the direction intersecting therewith. The example illustrated in FIGS. 10

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and 11 illustrates a configuration in which the unit structure 72 has a three-dimensional shape formed by connecting, by the side elements 73, atoms disposed at positions excluding vertices in a unit cell of a diamond structure. The unit structure 72 may have various structures, for example, a structure formed by connecting, by the side elements 73, vertices of a polyhedron such as a regular dodecahedron or a regular icosahedron.

Since the plurality of side elements 73 of the heat storage 71 are continuously connected to each other in a three-dimensional manner and provided in one piece, it is possible to reduce a dead volume which is a volume not involved in heat exchange in the housing 70, and to improve the heat storage performance of the regenerator 69.

Each side element 73 has a plate shape. Each side element 73 has a thickness of, for example, 0.1 mm or less, and is formed finely. Since it is necessary to repeat heat storage and heat release in a short time when exchanging heat energy with the working gas, each side element 73 is finely formed in this manner to reduce its heat capacity.

As illustrated in FIGS. 12 and 13, in each side element 73, a surface 731 having a maximum surface area is provided along the first direction. That is, each side element 73 is provided such that a projected area S2 as viewed from the second direction is larger than a projected area S1 as viewed from the first direction.

Since the projected area S1 in the first direction is small, the flow of the working gas is prevented from being hindered by the side elements 73, and the pressure loss at the time when the working gas flows in the first direction can be reduced. Further, since the projected area S2 in the second direction is large, the contact surface along the flow of the working gas is large, and the heat storage amount of the heat storage 71 can be increased. Accordingly, with the side elements 73, it is possible to achieve both a reduction in pressure loss and an improvement in heat exchange efficiency.

In each side element 73, in a cross section cut along the first direction, an end 732 on the heater 67 side and an end 733 on the cooler 68 side are provided in a curved shape. Specifically, in each side element 73, a cross section 73s cut along the first direction has an elliptical shape as illustrated in FIG. 12, and both ends of the major axis of the ellipse correspond to the ends 732 and 733. Accordingly, it is possible to reduce the pressure loss when the working gas collides with the ends 732 and 733 of each side element 73. The cross section of each side element 73 cut along the first direction is not limited to an elliptical shape, and may be any shape such as a rectangular shape. It is preferable that the ends 732 and 733 are provided in a curved shape or are inclined.

FIGS. 14 and 15 are a top view and a side view of the regenerator 69 in a state where the heat storage 71 is accommodated in the housing 70. In FIG. 15, only a portion in the first direction is illustrated. As illustrated in FIGS. 14 and 15, the heat storage 71 is uniformly disposed in the circumferential direction and the axial direction (first direction) in the housing 70, and as described above, since the housing 70 and the heat storage 71 are continuously connected to each other and provided in one piece, the strength of the regenerator 69 is high.

As illustrated in FIG. 14, as viewed from the first direction, the heat storage 71 includes pass-through portions 74 each of which penetrates from an end on the heater 67 side to an end on the cooler 68 side and that are regularly arranged. Specifically, each of the unit structures 72 has four through parts defined and formed in a lattice shape by the

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plurality of side elements **73** as viewed from the first direction, and the heat storage **71** having the pass-through portions **74** arranged in a lattice shape between the inner peripheral wall **61** and the outer peripheral wall **62** is formed by continuously providing a plurality of unit structures **72** in the first direction.

In this way, since the heat storage **71** has the pass-through portions **74** penetrating in the first direction, the working gas can smoothly pass through an internal space of the heat storage **71**. For example, when the heat storage is formed by stacking a plurality of metal meshes or the like as in the case of a heat storage in the related art, if the positions of the respective metal meshes are shifted, there is a possibility that the pass-through portions are not formed and clogging occurs inside the heat storage. In contrast, the pass-through portions **74** can be reliably provided in the heat storage **71** of the present embodiment.

In the heat storage **71**, the heat exchange amount with the working gas is larger at a position closer to the heater **67** in the first direction where the working gas has a high temperature and a position closer to the cooler **68** in the first direction where the working gas has a low temperature, and the heat exchange amount with the working gas is relatively small at a central position in the first direction.

Therefore, the side elements **73** of the heat storage **71** may have different shapes depending on the positions in the first direction. For example, the side elements **73** disposed at the central portion in the first direction may have a smaller projected area **S2** in the second direction than the side elements **73** disposed at both side portions of the heat storage **71** in the first direction.

In the heat exchanger **6** described above, the heater **67**, the cooler **68**, and the regenerator **69** are continuously connected to each other and provided in one piece. Specifically, the heater **67**, the cooler **68**, and the regenerator **69** are continuously connected to each other and provided in one piece in additive manufacturing using a metal powder material described later.

Since the heater, the cooler and the regenerator are provided in one piece, it is not necessary to separately provide the heater, the cooler, and the regenerator, to couple the heater, the cooler, and the regenerator by, for example, brazing or welding, or to provide the heater, the cooler, and the regenerator with flange portions to couple the heater, the cooler, and the regenerator by bolt fastening at the flange portions. Accordingly, it is possible to reduce the manufacturing time and the manufacturing cost of the heat exchanger **6**. In addition, since a member for coupling is unnecessary, the size and weight of the heat exchanger **6** can be reduced.

In a heat exchanger implemented by coupling the heater, the regenerator, and the cooler that are separately provided, leakage of the working gas may occur at the coupling portion. In contrast, in the present embodiment, since the heater, the regenerator, and the cooler are provided in one piece, leakage of working gas can be avoided.

[Method for Manufacturing Heat Exchanger]

The heat exchanger **6** according to the present embodiment is formed by additive manufacturing technology (hereinafter also referred to as AM technology) that can manufacture a component having a complicated three-dimensional shape by depositing and solidifying metal powder materials layer by layer. According to the AM technology, it is possible to manufacture a component having a fine and complicated three-dimensional shape, which is difficult to manufacture by a manufacturing method in the related art such as machining or casting.

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The heater **67** and the cooler **68** each have a flow channel (that is, the working gas flow channels **671** and **681**, the heat source fluid flow channel **672**, and the cooling fluid flow channel **682**) having a fine and complicated three-dimensional shape, and can be manufactured by the AM technology. In particular, the structure in which the spiral flow generation portion **55** is provided in the fine heat source fluid flow channel **672** is difficult to manufacture by a manufacturing method in the related art that does not use the AM technology.

A regenerator in the related art is not manufactured using the AM technology and is formed by stacking metal meshes or the like inside a housing. The regenerator **69** of the present embodiment can be manufactured by forming the housing **70** and the heat storage **71** in one piece using the AM technology. In particular, the heat storage **71** has a fine and complicated three-dimensional shape implemented by the AM technology, and is difficult to manufacture by a manufacturing method in the related art that does not use the AM technology.

As described, the heater **67**, the cooler **68**, and the regenerator **69** can be manufactured by the AM technology, and are formed in one piece by the AM technology in the present embodiment. Accordingly, as described above, a process (brazing or the like) for coupling the heater, the cooler and the regenerator is unnecessary, and the manufacturing time and the manufacturing cost of the heat exchanger **6** can be reduced.

The first cylinder **23** and the ceiling wall **27** thereof provided on the inner peripheral side of the heat exchanger **6**, the heat source fluid introduction chamber **80** provided above the heater **67** and the first cylinder **23**, and the heat source fluid discharge channel **51** provided on the outer peripheral side of the heat exchanger **6** are formed in one piece with the heat exchanger **6** by the AM technology. Specifically, as illustrated in FIG. 16, for example, a metal powder material is deposited and solidified layer by layer upward with the flange portion **25** of the first cylinder **23** as a bottom surface to perform additive manufacturing. In particular, the plurality of grooves **29** provided in the ceiling wall **27** of the first cylinder **23** have fine shapes, and are difficult to form by a manufacturing method in the related art that does not use the AM technology.

Accordingly, it is unnecessary to assemble the heat exchanger **6**, the first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51**, and the manufacturing time and the manufacturing cost of the Stirling engine **1** can be reduced. The first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51** may not necessarily be formed in one piece with the heat exchanger **6**. One or two of the first cylinder **23**, the heat source fluid introduction chamber **80**, and the heat source fluid discharge channel **51** may be formed in one piece with the heat exchanger **6**.

Further, since the fine flow channels and the heat storage **71** can be manufactured by the AM technology, the size of the heat exchanger **6** can be reduced. In the Stirling engine, when the heat exchanger is large, the dead volume of the space through which the working gas flows is large, and the compression ratio of the working gas decreases. Therefore, the heat efficiency and the output decrease. In the present embodiment, since the size of the heat exchanger **6** is reduced, the ratio of the dead volume to the volume of the heat exchanger **6** can be reduced, and the heat efficiency and the output of the Stirling engine **1** can be improved.

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In order to deposit the metal powder material at an inclination angle of less than a predetermined angle (for example, less than 45 degrees) with respect to a plane intersecting with a deposition direction when forming a component having a predetermined size by additive manufacturing, it is necessary to simultaneously form a support member that supports the inclined portion from below by additive manufacturing. When the support member is formed, it is necessary to remove the support member, for example, manually after component manufacturing.

In the present embodiment, in the heat exchanger 6, the first cylinder 23, the heat source fluid introduction chamber 80, and the heat source fluid discharge channel 51, an inclination angle equal to or greater than the predetermined angle (for example, 45 degrees) is appropriately set for a portion deposited on the upper side in an inclined manner, and formation of the support member is not necessary at the time of additive manufacturing. As illustrated in FIG. 16, for example, in the wall surface of the heat source fluid discharge channel 51 and the discharge portion 52 of the heat source fluid discharge channel 51, the inclination angle is set to 45 degrees for portions deposited on the upper side in an inclined manner. Accordingly, the formation of the support member can be avoided.

[Modification]

Although the Stirling engine 1 has been described as an example of the Stirling machine of the present invention in the embodiment described above, the present invention is not limited thereto. For example, the Stirling machine of the present invention may be a heat pump using a reverse Stirling cycle. Specifically, the heat pump serving as a Stirling machine may be configured such that the electric generator 11 of the Stirling engine 1 described above functions as a motor that inputs mechanical power to the piston crank mechanism 5 to cause the displacer piston 3 and the power piston 4 to reciprocate.

When the heat exchanger 6 is applied to the heat pump serving as a Stirling machine, the electric generator 11 functions as a motor, so that a fluid flowing through the heater 67 receives heat from the working gas and becomes a high-temperature fluid, and a fluid flowing through the cooler 68 has its heat absorbed by the working gas and becomes a low-temperature fluid. In the present embodiment, with a fine internal structure, the heat exchanger 6 has a reduced dead volume and high heat exchange performance, and thus the fluid can be heated to an ultra-high temperature or cooled to an extremely low temperature.

Although an embodiment of the present invention has been described above with reference to the accompanying drawings, it is needless to say that the present invention is not limited to the embodiment. It is apparent to those skilled in the art that various changes or modifications can be conceived within the scope described in the claims, and it is understood that the changes or modifications naturally fall within the technical scope of the present invention. In addition, the components in the above embodiment may be freely combined without departing from the gist of the invention.

For example, although the β -type Stirling engine 1 has been described as an example of a Stirling machine to which the heat exchanger 6 is applied in the embodiment described above, the present invention is not limited thereto. The heat exchanger 6 may be applied to a Stirling machine having another form, specifically, an α -type Stirling engine, a γ -type Stirling engine, or the like.

Further, although an example has been described in which the regenerator 69 is applied to the Stirling engine 1 that is

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an example of a Stirling machine in the embodiment described above, the present invention is not limited thereto. The regenerator 69 can be applied to various heat engines.

Although the side element 73 constituting the heat storage 71 has a plate shape in the embodiment described above, the present invention is not limited thereto. The side element 73 may be a linear body.

In the present description, at least the following matters are described. In the parentheses, corresponding components and the like in the above embodiment are illustrated as an example, but the present invention is not limited thereto.

(1) A heat exchanger (heat exchanger 6) for a Stirling machine (Stirling engine 1), the Stirling machine including an expansion chamber (expansion chamber 21) and a compression chamber (compression chamber 22), the heat exchanger including:

a heater (heater 67);

a regenerator (regenerator 69); and

a cooler (cooler 68), in which

the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid (working gas), and communicate the expansion chamber with the compression chamber,

the heater includes

a first flow channel (working gas flow channel 671) communicating with the expansion chamber and the regenerator and configured to allow the working fluid to flow therethrough, and

a second flow channel (heat source fluid flow channel 672) configured to allow a first fluid (heat source fluid) for exchanging heat with the working fluid to flow therethrough,

the cooler includes

a third flow channel (working gas flow channel 681) communicating with the compression chamber and the regenerator and configured to allow the working fluid to flow therethrough, and

a fourth flow channel (cooling fluid flow channel 682) configured to allow a second fluid (cooling fluid) for exchanging heat with the working fluid to flow therethrough, and

the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece.

According to (1), since the heater, the regenerator, and the cooler are provided in one piece by being continuously connected to each other, it is not necessary to perform brazing, bolt fastening at flange portions, or the like in order to couple the heater, the regenerator, and the cooler. Accordingly, it is possible to reduce the manufacturing time and the manufacturing cost of the heat exchanger. Further, occurrence of leakage of the working gas can be avoided.

(2) The heat exchanger for a Stirling machine according to (1), in which

the regenerator includes

a heat storage (heat storage 71) that is capable of storing heat of the working fluid, and

a housing (housing 70) that surrounds the heat storage, and

the heat storage and the housing are continuously connected to each other, provided in one piece.

According to (2), since the heat storage and the housing of the regenerator are provided in one piece by being continuously connected to each other, a complicated structure having high heat exchange efficiency can be adopted as the heat storage. Further, as compared with a regenerator in

which a plurality of metal meshes or the like are integrated into a housing, the number of assembly steps of the regenerator can be reduced.

(3) The heat exchanger for a Stirling machine according to (2), in which

the heat storage includes a plurality of linear bodies or plate-shaped bodies (side elements 73) that are formed in a three-dimensional manner and in one piece.

According to (3), since the heat storage includes a plurality of linear bodies or plate-shaped bodies that are formed in a three-dimensional manner and in one piece, it is possible to reduce a dead volume that is a volume not involved in heat exchange in the housing, and it is possible to improve the heat storage performance of the regenerator.

(4) The heat exchanger for a Stirling machine according to any one of (1) to (3), in which

the heater, the regenerator, and the cooler are an additive manufactured body formed by depositing a metal powder.

According to (4), the heater, the regenerator, and the cooler can be formed in one piece by additive manufacturing.

(5) A method for manufacturing the heat exchanger according to any one of (1) to (3), including:

forming the heater, the cooler, and the regenerator in one piece by performing additive manufacturing using a metal powder.

According to (5), the heater, the regenerator, and the cooler can be formed in one piece by additive manufacturing.

(6) A Stirling machine (Stirling engine 1) including:

a heat exchanger (heat exchanger 6) that includes a heater (heater 67), a regenerator (regenerator 69), and a cooler (cooler 68), which are connected in series in a flow direction of a working fluid (working gas); and

a cylinder (first cylinder 23) that includes an expansion chamber (expansion chamber 21) connected to the heater and a compression chamber (compression chamber 22) connected to the cooler, in which

the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger,

the heater, the regenerator, and the cooler are continuously connected to each other, provided in one piece, and

the heat exchanger and the cylinder are continuously connected to each other, provided in one piece.

According to (6), since the heater, the regenerator, and the cooler are provided in one piece by being continuously connected to each other, it is not necessary to perform brazing, bolt fastening at flange portions, or the like in order to couple the heater, the regenerator, and the cooler. In addition, since the heat exchanger and the cylinder are provided in one piece by being continuously connected to each other, a process of coupling the heat exchanger and the cylinder is also unnecessary. Thus, it is possible to reduce the manufacturing time and the manufacturing cost of the Stirling machine.

(7) The Stirling machine according to (6), in which the heat exchanger and the cylinder are an additive manufactured body formed by depositing a metal powder.

According to (7), the heat exchanger and the cylinder can be formed in one piece by additive manufacturing.

(8) The Stirling machine according to (6) or (7), further including:

a displacer piston (displacer piston 3) and a power piston (power piston 4) configured to reciprocate; and

a crankshaft (crankshaft 10) coupled to the displacer piston and the power piston, in which

the expansion chamber is a space defined by the displacer piston and the cylinder,

the compression chamber is a space defined by the displacer piston, the power piston, and the cylinder, and the crankshaft is connected to an electric motor or an electric generator (electric generator 11).

According to (8), a Stirling machine including a heat exchanger and a cylinder that are formed in one piece can be implemented.

What is claimed is:

1. A heat exchanger for a Stirling machine, the Stirling machine including an expansion chamber and a compression chamber, the heat exchanger comprising:

a heater;

a regenerator; and

a cooler, wherein

the heater, the regenerator, and the cooler are connected in series in a flow direction of a working fluid, and communicate the expansion chamber with the compression chamber,

the heater includes

a first flow channel communicating with the expansion chamber and the regenerator and configured to flow the working fluid to flow the working fluid therethrough, and

a second flow channel configured to flow a first fluid for exchanging heat with the working fluid to flow therethrough,

the cooler includes

a third flow channel communicating with the compression chamber and the regenerator and configured to flow the working fluid to flow therethrough, and

a fourth flow channel configured to flow a second fluid for exchanging heat with the working fluid to flow therethrough, and

the heater, the regenerator, and the cooler are disposed between a radially inner peripheral wall and a radially outer peripheral wall and the heater, the regenerator, and the cooler are continuously connected to each other in an axial direction of the radially inner peripheral wall and the radially outer peripheral wall and configured as one additively manufactured piece such that a separate coupling of the heater, the regenerator, and the cooler is not necessary to couple each of the heater, the regenerator, and the cooler to each other.

2. The heat exchanger for the Stirling machine according to claim 1, wherein

the regenerator includes

a heat storage that is capable of storing heat of the working fluid, and

a housing that surrounds the heat storage, and

the heat storage and the housing are continuously connected to each other, provided in one second piece.

3. The heat exchanger for the Stirling machine according to claim 2, wherein

the heat storage includes a plurality of linear bodies or a plurality of plate-shaped bodies, the plurality of linear bodies and the plurality of plate-shaped bodies are each formed in a three-dimensional manner and in one third piece.

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4. The heat exchanger for the Stirling machine according to claim 3, wherein

the expansion chamber and the compression chamber are formed inside a cylinder of the Stirling machine, and arranged in series along an axial direction of the cylinder interposing a piston therebetween, and

the radially inner peripheral wall and the radially outer peripheral wall extend in the axial direction of the cylinder along an outer peripheral surface of the cylinder.

5. The heat exchanger for the Stirling machine according to claim 1, wherein

the heater, the regenerator, and the cooler are an additive manufactured body formed by depositing a metal powder.

6. A method for manufacturing the heat exchanger according to claim 1, comprising:

forming the heater, the cooler, and the regenerator in the one piece by performing additive manufacturing using a metal powder.

7. A Stirling machine comprising:

the heat exchanger of claim 1; and

a cylinder including an expansion chamber connected to the heater and a compression chamber connected to the cooler, wherein

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the Stirling machine is configured to perform a Stirling cycle to convert heat into mechanical power by repeatedly changing volumes of the expansion chamber and the compression chamber while circulating the working fluid between the expansion chamber and the compression chamber via the heat exchanger.

8. The Stirling machine according to claim 7, wherein the heat exchanger and the cylinder are an additive manufactured body formed by depositing a metal powder.

9. The Stirling machine according to claim 7, further comprising:

a displacer piston and a power piston configured to reciprocate; and

a crankshaft coupled to the displacer piston and the power piston, wherein

the expansion chamber is a space defined by the displacer piston and the cylinder,

the compression chamber is a space defined by the displacer piston, the power piston, and the cylinder, and the crankshaft is connected to an electric motor or an electric generator.

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